Observers Who Read Their Universe into Existence: Finite Consensus and the Reconstruction of Standard Physics

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Abstract

Observer Patch Holography (OPH) proposes that the Universe is what its observers agree on. Its target is an observer-based computation whose recorded experiences reproduce our Universe in every physical respect. We specify a simple architecture: small observers on a twelve-port carrier keep local records, compare them with their neighbours and repair disagreements with one local rule. We study this native carrier together with explicitly declared field extensions. Its elementary slow-response eigenspace is three-dimensional. Exact native channels yield interacting exclusion records; declared coherent services supply quantum measurement identities from the Born rule to the Tsirelson bound. Under complete-current premises the gauge Lie algebra is that of the Standard Model; a declared matter representation carries the observed generation charges. A generation register yields three charged-family energies with ratios 206.770316 and 3477.4728, within 0.001 and 0.0032 per cent of measurement in the retrospective comparison. A positive native strong action with seven declared inputs fits nucleon and pion energies and deuteron binding. A common finite program transports charged records and integer electric flux, preserves every Gauss constraint, and runs two clock observations through the same field evolution with explicit tracking services. A separately declared cubic routing yields a controlled isotropic scalar dispersion. Conditional on a common realized null balance, tensor completion gives the Einstein equation. The public comparison ledger separates exact finite identities, numerical comparisons, calibration targets and open physical identifications, and retains the tests that can refute each construction.

Keywords: foundations of physics; Observer Patch Holography; finite consensus; self-reference; quantum records; gauge structure; particle physics; Einstein dynamics.

1 Introduction

Mathematics describes structures that may be physically real. An equation for a wave or a fixed point states relations; a physical realization gives those relations observable content. The existence question asks whether the whole process could admit an explanation of this kind. A brute initial fact ends explanation at an unexplained starting point. A chain of prior causes moves the same

question to an earlier link. Neither option is a logical contradiction, but neither explains the whole on its own. OPH asks whether the complete history can close on itself instead.

The inversion is that a description of the world is itself an event in the world. Observers infer rules from public records, construct a process that implements those rules, and belong to the history that the process describes. The self-reference has content only if one specified history actually satisfies this return condition and reproduces the observations made within it. This is the governing question of the paper. Consistency constrains the construction; it does not make an arbitrary equation into an existent universe.

1.1 Mission

The mission of OPH is a consistent and complete, formal and rigorous theory of reality. Its proposed complete history has no external starting point: it closes on itself. A successful theory must meet four conditions.

1. **Consistent.** The loop closes. Its fixed point exists, is constructed explicitly and is machine-checked, and a theory with a model contains no contradiction.

2. **Explicit.** The complete recovered specification fixes its consequences. Every design choice and calibration is declared; quantities claimed as independent predictions are not inserted through those inputs.

3. **Complete**, in the sense Einstein, Podolsky and Rosen gave the word [27]. Every element of physical reality has a counterpart in the theory, and every measured quantity is derived.

4. **Correct.** Derived values agree with measurement, and new predictions are frozen before they are tested.

The mission ranges from the existence of an inhabited history and the possibility of experience to the number of spatial directions, quantum statistics, forces, particles, gravity and cosmology. This range specifies what the proposed account must explain; it does not turn a finite model or a conditional theorem into a derivation of every item. In particular, a three-dimensional carrier response needs a physical position map, and an operational observer needs an additional account before any claim about phenomenal qualities follows. Table 3 gives these distinctions alongside the physical results.

OPH does not claim to be the only theory with these properties. The four conditions organize this paper. Consistency is tested through the closure hypothesis and the executable model (Section 2) and through the machine-checked results of Section 14. Specification and parameter ancestry are tested through the closure coordinates of Section 11.4. Completeness is the ledger of realized results and physical obligations in Section 12. Correctness is decided by the frozen predictions and tests of Section 13.

1.2 Consistency through local repair

Local repair is the engine of the proposal. A finite observer has no global view. It holds a patch of local state, reads the boundary shared with neighbors, retains records and repairs disagreements while protecting what has been recorded. A fact becomes public when every authorized comparison preserves it. The finite consensus theorem gives such records a canonical normal form independent of the repair schedule. Observer Patch Holography (OPH) asks whether the world we measure can be reconstructed from this kind of compatible local access. Quantum probability, Lorentzian

kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [28, 29, 30, 64, 71]; OPH tests whether one observer architecture can carry their public readouts. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret. In this interpretation, spacetime geometries and quantum field theories describe relations among compatible records; the proposal assumes no prior observer-independent continuum.

The twelve-port carrier makes this idea concrete. Optimal readout of a three-dimensional response selects its incidence, and its normalized repair response has a three-dimensional slow quotient, a candidate carrier for spatial position. Under the further premise that a faithful complete reversible current implements the carrier's rechartings internally, its Lie algebra has type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Three directions and the Standard Model gauge type thus arise as different constraints on one boundary network's response. The defects that repair leaves behind carry exactly the six centre classes of the Standard Model particles, and a generation register in the same law gives the charged-lepton mass ratios. Identifying those structures with physical position and laboratory currents is a further test.

At continuum scale, consistency links matter and geometry. If one history supplies conserved stress and makes every local null direction obey the same curvature–stress balance, tensor completion yields the Einstein equation up to a cosmological constant. Obtaining that balance, the entropy response and a common physical geometry from the finite source is a separate requirement. OPH's unifying hypothesis is that the full dynamics of nature are the conditions under which one self-reading process keeps its records coherent through local repair.

The intended argument runs from specified repair operations to compatible joint records, and from those records to a common operational description of space, time and matter. A position is what the spatial readouts locate; a duration is what calibrated clocks compare; a particle is a persistent excitation with the observed transport, charges and interactions. The target is one such description with three spatial directions and local time, including its quantum fields and gravitational response. The finite constructions below establish parts of this chain; the common physical $3 + 1$ limit is the remaining conclusion to be demonstrated, not an input hidden in the word "observer."

OPH makes one proposal about the whole, closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover its specification and realize it (Section 2). The executable-model hypothesis is the concrete and testable form of that proposal: one specified local repair and memory process must supply records, clocks, fields, matter, energy and responsive geometry together. This paper presents OPH as that single theory, organized by its dependencies: each layer states a hypothesis, proves finite theorems on displayed premises, and passes a conclusion upward only when the executable model realizes the premise on the same process.

1.3 Scope and order of the argument

The program has four logically different levels. Closure proposes that an inhabited history contains a recoverable account of its own operations. A source specification states the carrier, observer access, repair, record and state laws. Finite theorems and executable certificates determine what that specification does. Physical realization asks whether one such source has the observed fields, particles, clocks and responsive geometry. The implications run in that order; Table 1 states the result at each level.

The finite results make OPH a concrete reconstruction program. Their composition into the observed universe is a separate mathematical and empirical claim. The specialist papers give the

Table 1: The OPH program and the status of its main implications.

Level	Result on its stated domain	Physical requirement
Closure	Finite consensus has an exact fixed-point packet theorem; a universe-level self-recovering history is proposed	A specified history and realization map satisfying the closure condition
Records and carrier	Canonical public records, conditional quantum identities, finite entropy laws of repair and an exact rank-three slow response	A common physical position, clock, energy and preparation map
Gauge and matter	Complete-response premises force the Standard Model Lie type; protected defects carry exactly the particle centre classes; port phases give the global group; a generation register gives the charged-lepton ratios; a declared coupling chain and lattice ensembles give the electroweak crossover, colour deconfinement and hadron binding	An interacting chiral field with the observed absolute masses, quark flavours and mixing; a derivation of the chain
Gravity and cosmology	Null tomography and entropy premises imply the Einstein equation; finite capacity and pixel equations give conditional relations	The same source must supply conserved stress, responsive geometry, vacuum, scales and prospective observational matches

longer proofs, source executions and failed attachments [13, 12, 10, 72].

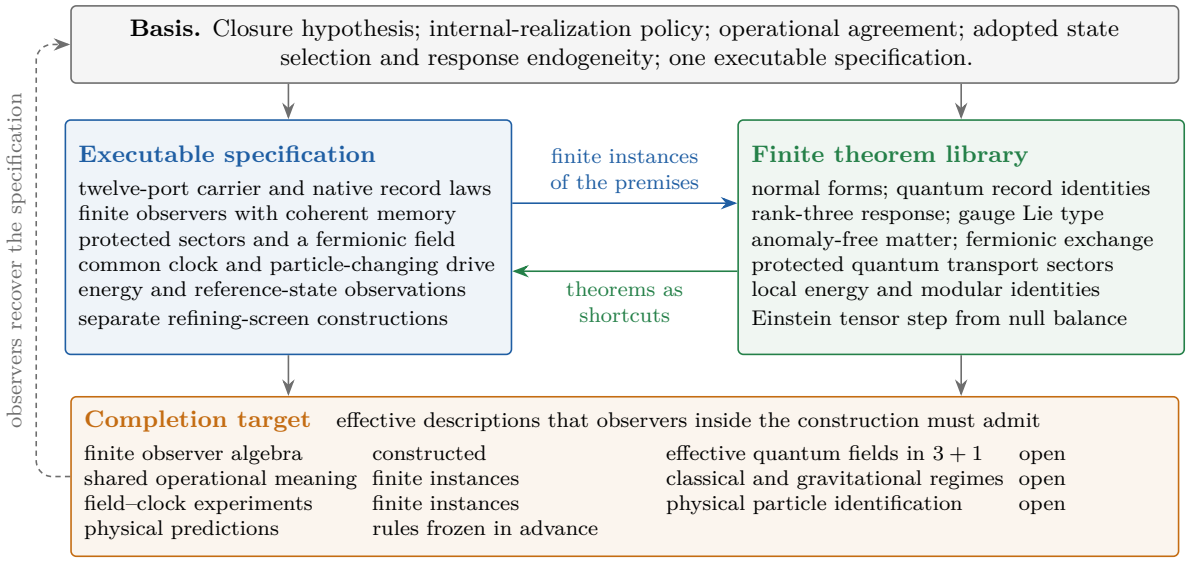


Figure 1: The proposed chain from protected records to physical laws. The finite theorems and source executions constrain one specification; physical conclusions require compatible realization maps. The dashed arrow is the closure proposal: observers recover the specification from their records.

Finite theorems, complete source executions and physical interpretations answer different ques-

tions. We state each result's domain and supplied inputs where it is used; the detailed claim classes and verification methods are collected in Section 14.

"Source-executed" means that an operation is compiled and run using the declared native preparation, repair, memory and readout services, with its complete outputs and resources accounted. "Source-only" concerns the provenance of its parameters: no empirical physical input occurs anywhere in their dependency chain. These are different properties. A source-executed field may use empirically calibrated Yukawas, mixing parameters or clock units; reproducing those inputs verifies the implementation, not an independent prediction.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

The paper gives one mathematical representative of each step: the native repair rule, the consensus normalizer, the spatial and gauge reductions, the finite matter attachment and the Einstein tensor implication. Its status tables and prospective tests specify the physical conclusions. The companion construction and component papers carry implementation contracts and longer proofs [72, 1, 5, 6, 7, 9, 12, 10, 8, 14, 15, 26, 3].

2 Closure and the source-specification hypothesis

This section states the interpretation and acceptance criterion: closure, the internal-realization policy, the additional state-selection and response-endogeneity principles, and the agreement condition, then explains why closure is tested through an executable model.

2.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R\big(D(\mathfrak{U})\big) \;\cong\; \mathfrak{U}, \tag{1}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

Within this hypothesis, no external computer runs the history and no additional time exists in which the whole is played. Host runtime used to check a model is part of our method of investigation, not an extra time coordinate of the universe described by that model. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration.

Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x+1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x+2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, consistency and the identification of our universe with a fixed point are questions the equation leaves open. Self-reference alone guarantees neither consistency nor mathematical completeness. Uniqueness is an additional question when it is claimed; it is not a prerequisite for a specified construction to work and match observations. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms [7]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [9].

2.2 No outside

A self-contained realization must account internally for its operations and information access. This motivates a justification policy and two further constitutive principles; it does not logically derive those principles.

Justification policy. A premise is justified operationally when observers recover it and the specification implementing it runs and reproduces their measurements. The carrier, observer algebra, repair law and closure coordinates must meet that test. Their specification can be reconstructed from records or proposed as a design. In particular, using a twelve-port carrier does not require a theorem that every possible realization must have twelve ports. A construction choice needs no uniqueness theorem unless uniqueness is claimed. Closure neither exempts a candidate from empirical tests nor proves its antecedents jointly realizable.

State-selection principle. On the declared feasible family, reference, observer cover and positive weights, adopt the information projection minimizing $\sum_p w_p D(\rho_p \| \tau_p)$. This is premise A3, motivated by least-informative inference but separately adopted when used as a physical selection law. The closure equation supplies no entropy functional or reference; even its identity-map specialization fixes both uniform and nonuniform states. Under A3, every minimizing local density has the union of feasible supports, and a feasible failure witness $b > 0$ gives an explicit selected-probability floor [10].

Response-endogeneity principle. The complete-response branch additionally requires each proper carrier recharting to be implemented by the connected response group, up to a unitary centralizer acting trivially on the response algebra. This is the endogeneity clause of A2, not a consequence of internal implementation alone. For example, twelve diagonal phase generators on $\mathbb{C}^{12}$ admit internal port permutations in $\mathrm{U}(1)^{12} \rtimes A_5$. These permutations act nontrivially on the abelian current algebra, whereas its inner automorphisms are trivial. This closed operation system fails the stronger endogeneity premise; it shows why that premise must be supplied. Theorem 8.1 uses it to select $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under the stated complete-response hypotheses.

2.3 Agreement and the meaning of "our"

The justification policy fixes the operational test. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 4.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: it requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. Agreement here is compatibility under specified comparisons and interventions, not a vote among observers or assent to a convenient interpretation. Its content includes the joint probabilities and the response to changed preparations. Calling space, time and matter operational descriptions does not make these constraints optional: once established by compatible records, they are public physical facts. The equations describe those relations and do not by themselves assert an additional material substrate beyond the specified process. The theorem library calls the interpretation clause of this condition A2. The basis reference distinguishes the closure hypothesis, agreement, adopted selection/endogeneity principles and executable specification, with the library's premise statements and their dictionary [103].

2.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. Closure states the internal realization contract; the adopted principles and specification then carry their own mathematical and empirical tests. The order of exposition is a logical entry point, not an event before the universe or a privileged temporal beginning of the self-referential loop. A pair of mutually determining equations such as $x = (y + 2)/2$, $y = (x + 2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two.

The closure equation relates parts of one timeless object; it posits no backward signal in an external time coordinate. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. In the terminating consensus theorems, repair descent supplies a well-founded order on the admitted updates for proving termination and comparing schedules; it is neither an observer's experienced time nor a universe-level clock.

What is reconstructed and what is constructed are two roles within this proposal. The observer who infers a law and the hardware that realizes it both belong to the candidate history. Finding that specification in a laboratory or checking it by computation occurs in the investigator's record order; neither supplies a temporal origin outside the history. If several specified closures satisfy the same tests, the results warrant that observational class without a uniqueness claim.

2.5 From closure to a testable specification

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence. A playback can match every frame of one history and fail this test as soon as a retained state is changed. The controls of Section 6 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

Closure alone does not imply that a particular carrier, record law or initial state yields Standard Model physics. The executable-model hypothesis is the stronger claim that one simple, fully specified law does, and it is the form in which closure can fail. Its carrier architecture, refinement and composition, record instruments, memory law, scheduler, preparations and available controls are stated before testing. A failed specified law rejects that law and its tested interpretation; it reaches the wider proposal only when the rejected choice is proved necessary and exclusive. Changing the law, memory, state or decoder defines a new candidate.

2.6 Specification and theorem premises

The architecture, record instruments, memory law, scheduler, preparation services and allowed interventions form a declared specification. Observers may infer or design these ingredients; mathematics determines their consequences, and comparison determines whether the resulting process matches observed physics. Declaring a primitive fixes the model under study. Its consequences require proofs, and its physical adequacy requires tests; a specified input must remain distinguishable from a consequence derived from other inputs. The carrier, memory and control laws need not be uniquely selected by closure to be legitimate inputs. Their implementation and resource use must, however, belong to the process being studied.

Three premise families recur. A1 specifies the carrier, federation and response structure. A2 adds agreement on shared records and inner implementation of the complete response algebra. A3 adopts a least-informative state-selection principle on the stated feasible family. Closure alone proves neither the entropy minimizer nor the inner implementation assumption. Later theorems state which of these inputs they use and which additional hypotheses their conclusions require.

Quantum executions below use finite-dimensional complex Hilbert spaces, positive normalized density operators and completely positive instruments whose complete effects sum to the identity. These are explicit quantum premises, not consequences of classical normal-form uniqueness. A faithful reference is a strictly positive density operator $\rho > 0$; its modular flow is $\sigma_s(A) = \rho^{is} A \rho^{-is}$. The dimensionless parameter s requires a separate implemented calibration before it is read as duration. For fermionic fields the canonical anticommutation relations (CAR) are $\{c_i, c_j^\dagger\} = \delta_{ij} I$ and $\{c_i, c_j\} = 0$; the declared Fock representation fixes the exchange signs used later.

The physical interpretation also fixes a domain: one prepared history, a family of interventions, or every allowed input state. These are inequivalent requirements. Complete quantum instruments must retain the reference systems and all outcomes needed for the chosen domain; a record history need not assign simultaneous values to unperformed incompatible measurements. Likewise, a stationary probability law, an instantaneous repair fixed point and the closure of an entire history are distinct. Host runtime, repair counts, modular parameters and measured duration are related only by an implemented clock calibration. The operational observer below is defined by access and record interfaces, without an additional claim about consciousness.

3 A finite observer specification

The model rests on two objects: an operational observer and a twelve-port carrier with its native repair law. We define their interfaces and update rules before constructing the observer that runs on them.

3.1 Patches, records and repair

Definition 3.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 3.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

3.2 The twelve-port carrier

Each local carrier has twelve boundary ports, thirty seams and twenty oriented triangular faces. The incidence has a selection theorem that uses no topological template, Euler count, group action or preassigned port count.

Theorem 3.3 (Icosahedral incidence from optimal readout)**.** *In a real three-dimensional response space, let unit vectors u_i read symmetric operators through $\mathrm{tr}(P_i X)$, $P_i = u_i u_i^{\mathsf{T}}$, with positive weights $\sum_i w_i = 1$, and put $M = \sum_i w_i |P_i\rangle\langle P_i|$. Informational completeness requires at least six distinct channels, and every choice obeys $\lambda_{\min}(M) \le 2/15$. With six channels, equality holds exactly for equal weights and the six icosahedral axes, up to an orthogonal frame. Exposing both signs gives twelve ports, positive overlap gives degree five and thirty edges, and the three-cliques give twenty faces. A deficit of at most 10^{-7} forces the same incidence, without forcing exact angles.*

Proof. Sym(3) has dimension six and each term of M has rank one. Each $P_i - I/3$ has squared norm $2/3$, so the trace-free compression of M has trace $2/3$ on five dimensions, which gives the bound. Equality forces $M(X) = (2X + \mathrm{tr}(X)I)/15$. At six channels the matrix F with columns $\sqrt{w_i} P_i$ is invertible and $F^{\mathsf{T}} M^{-1} F = I$: its diagonal gives $w_i = 1/6$ and its off-diagonal gives $(u_i \cdot u_j)^2 = 1/5$. The Gram matrix $I + S/\sqrt{5}$ then has a sign matrix with $S^2 = 5I$, whose unique switching class fixes the geometry [8]. □

The response dimension, the quadratic readout, both objectives, inverse ports and positive-overlap edges are premises, and a physical metric requires operational reconstruction. Carriers federate through typed seam algebras and coherent triple overlaps into a nerve with a degree-one bridge to an oriented sphere. Native current records recover the local Gram; executed cross-carrier routing uses supplied cell centres and tangent frames. A1 in the theorem library declares this carrier, federation and response layer.

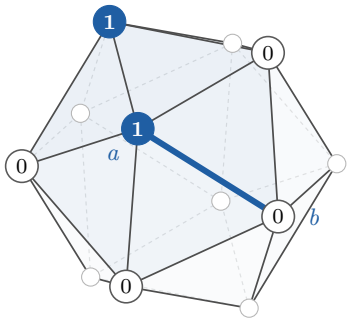
Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{2}$$

On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real

representations of real type; the projectors, characters and irreducibility certificates are machine-checked [59, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates. The native repair law of Section 3.3 also moves load to the floor and ceiling of the endpoint mean, but at difference one it waits or swaps by its coin, so the native process keeps moving while its quadratic cost stays fixed.

3.3 The native repair law



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 2: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$ (Figure 2), the native repair law moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(e_b - e_a), \tag{3}$$

so load is conserved and the signed current j is the native observation. Conservation, one-step balance, a tie-break that reads no load and endpoint symmetry characterize this rule; on the total-two carrier they fix only the tie-break coin. Its quantum version is a record law, and seam locality, the classical law and the declared local number-phase and endpoint-swap covariance, a local phase symmetry of the repair channel, leave a convex family of them, each a table of five coherence multipliers [74]. The two laws used below have the same load update (3) and differ in which information they record. Write $p_e(x) = (x_a + x_b) \bmod 2$ for the parity of the seam's total load. Law C records the pair (c, j); law E records (c, j, p_e). Their Kraus operators are

$$L^C_{c,j} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \tag{4}$$

$$L^E_{c,j,p} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j,\ p_e(x)=p} |y(x,c)\rangle\langle x|. \tag{5}$$

Each complete instrument preserves total probability. E exposes an extra parity record and therefore removes coherence between the corresponding parity classes when the record is ignored. It changes no classical load or current probability. C is the convention used for the primary construction; E provides a specified alternative and control. A current-only coherent implementation gives the most coherent member C_0; adding a coin readout gives C, and adding seam parity gives E. Classically forgetting a written coin is not the coherent C_0 implementation. E and the seam-total recorder are the two idempotent members of the classified family. Under a common definite-current record interface all members share every classical current statistic. One coherent interference experiment reads the multiplier g that separates C, E and the most coherent member (Section 13), and nine source-executed measurements read every real coordinate of the table (Section 6). The unread channels have a source-independent spectral gap $(5-\sqrt{5})/60$ on the binary shell. Every admissible coherence table has the uniform binary state as its unique stationary state, and the rank-three load readings of Theorem 5.1 attain the slow rate; equilibrium therefore does not select C or E [74, 75]. Section 4.4 gives the entropy laws of the same repair.

The local operator $H_{\text{native}} = \sum_e (I - \Phi_e)$, with Φ_e an unread seam channel, is a positive Liouville operator on observables. It is not a Hamiltonian on carrier states: the executed channel attenuates coherence and cannot be unitary conjugation on this carrier. The separate field constructions must supply their own quantum evolution and physical energy interpretation [87, 73].

4 Public records and completed-record identities

The repair mathematics establishes when local agreement reaches global consistency and when nonzero cycle holonomy leaves a residue that local repair cannot remove. Two observers who share a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records satisfy the familiar quantum identities on the declared complex algebra and state domain below.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\text{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 2. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [5, 7].

4.1 Observable normal forms

Definition 4.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 4.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ that preserves B on its C-valued outputs, fixes every point of C, is constant on each B-fiber, and returns $\bot$ exactly on fibers with no consistent extension;

If $B(Q) \subseteq B(C)$, the map is a total idempotent retraction $Q \to C$.

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [5].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q + 1$ attempts, a bounded-waste upper horizon of $(q + 1)$ times the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n + 1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [5, 7].

4.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 4.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 4.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 4.3, so that it has the one-step diamond property, strictly decreases the quadratic descent functional V of Section 3, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 4.3 makes it locally confluent, so Newman's lemma [52] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

Repair also has a residue that it cannot remove. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, and no repair of the vertex assignments eliminates it while the offsets are fixed [7]. The protected defects that this frustration leaves behind are the particle families of Section 8.6.

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [3]. The level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. The excess-energy contraction per sweep lies between 0.81 and 0.83 across rungs five to nine, and these finite readings prove no all-size mixing or termination-time bound. Stationary continuation adds a constructive

fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [8].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations, and tetrahedral and octahedral controls recover their distinct specifications [3]. Complete federation gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

4.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional complex C^*-algebra with a positive normalized state and completely positive instruments. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public and does not by itself select complex quantum theory over every alternative probability model.

Theorem 4.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\, \rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{6}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [33, 34]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1, 1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [35], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [6]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \operatorname{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [31, 32]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. For a complete Kraus instrument, basis survival and one coherent survival probe per nontrivial outcome block select both the effects and the Lüders map. Approximate tests bound complete outputs with arbitrary references; trusted preparation and readback are premises [38, 6]. The same

surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [36, 37].

Native repair histories construct a functional of this form on a full matrix algebra. The complete-attempt transfer P of the 66-state carrier comes from all 3960 endpoint branches and obeys $2I/3 \le P \le I$. The twelve occupation fields and connected transfer paths generate all 4356 matrix units, scaled closed-history counts determine their moments, and the unique positive density is $\rho_L = P^L / \operatorname{Tr} P^L$. Its canonical complex-linear extension is additive and positive, so the Born theorem applies on this reconstructed algebra. A current-only return census implements the required configuration comparison, with at most 11398 attempts per experiment at the stated settings and systematic error at most 3.76596×10^{-6} per reconstructed effect statistic. Finite polynomials reconstruct ordered modular correlations with the common generator $-\log P$, with bias below 4.302×10^{-10} in the displayed three-time example. The entries of this functional are classical endpoint probabilities divided by the return probability, every connected positive-definite symmetric stochastic matrix gives the same result, and native C and E evolution converges to $I/66$ without preparing it. Direct native acceptance effects are diagonal. Complex tomography uses supplied coherent services, and the operational bridge requires equal-effect probability independence and disjoint common implementations of coexistent pairs [6, 9].

4.4 Entropy and thermodynamic laws from repair

For a conditional-resampling repair law with a stated reference and energy calibration, finite theorems give the second law, a Clausius inequality, fluctuation relations and a Landauer bound [9]. Native repair realizes this structure directly. On binary populations, averaging the two fair-coin outcomes of a seam gives $P_e = (I + \mathrm{swap}_e)/2$, exactly uniform resampling conditional on the outside-seam loads. The connected seam ensemble has one uniform stationary reference, which uniquely maximizes entropy in its sector, and

$$S(pP_e) - S(p) = D(p \| pP_e) \ge 0$$

for the unread population; fixed seam words telescope this identity. Retaining the complete endpoint and coin/current tape gives $H(X_n, J) = H(X_0) + n \log 2$, so the marginal entropy increase loses nothing from the complete record. The modular energy of this branch is constant, which leaves its temperature undefined.

A native memory refresh supplies a nonconstant calibration. A stopped coin-prefix program records 0, 10 or 11 after one, two or two attempts, with weights $(1/2, 1/4, 1/4)$ that ignore the hidden configuration. The stored-prefix cost $E = n \log 2$ gives weights e^{-E}, and overwriting the memory with a fresh prefix is exactly its global heat bath in at most two native attempts. Agreement of b independent prefixes gives the Gibbs family of the same cost at integer multiplier b. Keeping the old memory after disagreement gives a lazy reversible kernel with that stationary reference, bounded retries keep every rejection outcome with an explicit geometric error, and almost-sure retry gives the exact heat bath at finite expected cost. A Lean instance constructs the repair and calibration data from these counts and applies the finite four-law core, with gap $\log 2$ and a unique ground label. That core holds for every full-support law with $E = -\log(\mathrm{ref})$ and formalizes no contact relation, the fair coin is a declared input, and record cost is distinct from laboratory heat, temperature and total implementation work [9].

Pressure needs a strain law. The family $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ has the same fixed-volume energy histories and $p/\rho = \alpha$, so consensus about identical fixed-volume records cannot select the strain derivative. Three geometrically spaced energy readings identify it within the held-state family

$C + A(V/V_0)^{-\alpha}$, with exact intervals propagating meter error, and a complete two-attempt native E census reads independently calibrated charge-diagonal forces. An additive energy offset changes p/ρ while leaving pressure unchanged. Volume and energy calibration are supplied [9].

5 Three spatial directions

The carrier incidence fixes the dimension of the carrier's own position readback: the slow band is the three-dimensional irreducible representation of the icosahedral group, so the number three is a property of the committed carrier.

Theorem 5.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1 - (5 - \sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (2) carried by the port position vectors. The same number $(5 - \sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 3.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [73]. Reading the completion as physical position consumes the realization map (M1) of Section 12.1 [1, 8]. The rank-three readout belongs to one carrier's response. When carriers join into the refining spherical screen of Section 7.1, the continuum form of their repair history has spectral dimension two, the dimension of the screen, and bounded-energy families lose every within-carrier component, this band included, at the present full-port energy scaling; a three-dimensional bulk reading of the screen requires observation maps that the native energy does not supply.

6 The source specification in operation

The finite theory has to run as a process that observers can access. Its specified carrier, complete instruments, scheduler, memory, controls and preparations define that process. Source executions check that the same operations produce the claimed conditional records, including the outcomes

that an observer does not see. The construction paper gives the full implementation and certificates [72, 73].

Records affect later operations. A two-event observer writes the first seam current, selects its next seam from that record and keeps the second outcome. Exact enumeration gives final port-zero occupation 35/264, versus 1/6 for a disconnected or frequency-matched shuffled selector. The comparison isolates the causal role of the retained record within this specified rule. It does not select that rule as the one realized by nature [76].

Observers share one source. Two observers with overlapping port access read a continuing native history, each with its own incident-event clock. Their decoded values at the common port agree on all 5,768 retained output words, including waits and timeouts; a separate transfer recursion reproduces the same outputs exactly. Federated-screen executions and a locality estimate carry complete root-observer outputs across declared larger regions [82, 88]. These are finite agreement and locality results. Physical position and duration require additional realization maps.

Quantum instruments and memory. The C and E instruments implement the same classical seam update while recording different information about a quantum input. Interference can distinguish them, and a finite measurement menu identifies the coordinates of their declared record-law family [86]. A fixed 570-attempt native preparation accepts arbitrary total-two inputs, including entangled references, with error below 3.72×10^{-8}. Ignoring its warm-up tape adds that error once to all 8,195 records of the capped phase/work continuation; coherent-compiler error is retained separately [9]. On its supplied work register, gauge flux, a charged pair and scalar sites have non-commuting evolution and record feedback. A smooth four-read continuation has complete $O(1/d)$ error, and active recovery protects a specified logical qubit [77, 78, 79, 81].

The source executions establish that declared operations compose and that records can control later field and repair actions. They do not derive an autonomous observer population, a physical clock or an interacting relativistic continuum from the finite tests. The physical status of those joins is stated in Section 12.

6.1 Measurement and public records

A private outcome becomes public when authorized comparisons preserve it through repair to a common normal form. Event projectors, weights and conditioning then act on the completed algebra-state surface of Section 4.3. The running specification demonstrates this transition with full conditional instruments and untouched reference systems. Quantum identities use the supplied complex algebra-state representation; the finite measurement protocols verify its declared implementation on their stated domains [72, 80].

7 Fields, clocks and causal order

The same carrier selects its field action. Write D and C for the seam-by-port gradient and face-by-seam curvature, so that $CD = 0$.

Theorem 7.1 (Local Maxwell selection)**.** *A real quadratic seam action $S(A) = \text{const} + \ell^{\mathsf{T}} A + \frac{1}{2} A^{\mathsf{T}} H A$ that is gauge invariant, has off-diagonal Hessian entries only between seams of one face, and is covariant under the sixty carrier rotations is exactly $S = \text{const} + \frac{\kappa}{2} \|CA\|^2$. For positive*

weights ϵ, β, a real linear autonomous evolution with $\dot{B} = -CE$, conserved $(\epsilon\|E\|^2 + \beta\|B\|^2)/2$ and reversal with E even and B odd is exactly $\dot{E} = (\beta/\epsilon)C^{\mathsf{T}}B$ on closed flux.

Proof. Gauge invariance is $HD = 0$ and $D^{\mathsf{T}}\ell = 0$. At an edge of a face and its opposite vertex, locality permits exactly the other two edges in that entry of HD, and the signs of $CD = 0$ propagate one weight w_f over the face. Subtracting $\sum_f w_f C_f^{\mathsf{T}} C_f$ leaves a diagonal matrix annihilating D, hence zero. Transitivity on faces equalizes the weights, and the sum of the sixty signed seam actions vanishes, which removes ℓ. For evolution, conservation makes the electric block skew and fixes the cross block on closed flux, and reversal removes the skew block [8]. □

Exact rank computations and countercontrols verify the classification. The same form is the leading response of a smooth conservative field law, with a uniform finite-time bound $\|x(t) - e^{tG}x_0\| \le (L/2)J_0 r^2|t|$ and no linear ansatz; exact nonlinear Gauss conservation needs an additional condition. On a supplied Euclidean domain, the complete equal-weight seam symbol Λ_a obeys $0 \le r^2 - \Lambda_a(k) \le a^2 r^4/20$ at $r = |k|$, and its reversible curl-pair group converges to the local Maxwell group with $\|U_a(t,k) - U_0(t,k)\| \le \min\{2, |t|a^2 r^3/20\}$. The premises, field identification, coefficients and clock are inputs, and native repair does not select them.

The action principle itself has a source form.

Theorem 7.2 (Derived action of a source chain)**.** *Let P be a strictly positive row-stochastic kernel on a finite state set Ω and π a strictly positive initial law. On paths with n transitions put $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$ and $R(\gamma) = \pi(\gamma_0)|\Omega|^{-n}$. The Markov path law is the exponential tilt of R by $S_{\log}$ at multiplier one, with partition constant $|\Omega|^{-n}$, and a pair (S', λ') reproduces it exactly when $\lambda' S' = S_{\log} + c$ for a constant c.*

Exponentiating the negated action recovers the transition product, and row normalization telescopes the partition sum; the uniqueness clause follows by comparing logarithms at a common path, and the statements are checked in Lean [9]. The committed source chain instantiates the theorem exactly. Its bare repair count reproduces the path law at no multiplier, which makes the log-transition action the derived object. Physical field actions are a further attachment.

7.1 The native screen operator and a conditional spacetime field

On the geodesic screen of level l, let $V_l = 240 \cdot 4^l$ and prepare the uniform canonical state with total load $V_l/3$. Independent uniform seam selection and fair native repair define one complete occupation history. Write $\chi_l = q_l(V_l - q_l)/[V_l(V_l - 1)]$ for $q_l = V_l/3$.

Proposition 7.3 (Native continuum operator)**.** *Along a common refinement subsequence, the native cell Green operators converge in norm to a compact positive injective operator G on the mean-zero observation space. After restoring the constant mode, its inverse A generates a regular strongly local conservative form, with domain H^1, and*

$$n/49920 \le \lambda_n(A) \le 3n/2.$$

The complete density readouts

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\, 16^l \tau \rfloor}(p) - 1/3]}{\sqrt{V_l \chi_l}}$$

converge jointly at finite sets of times to a stationary reflection-positive Gaussian field with covariance $\langle f, e^{-|\tau - u|A} g\rangle$. The admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. With the constant mode restored,

$$\|\mathbf{1}_F e^{-\tau(A \oplus 0)} \mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)},$$

and the cosine propagator vanishes for $|t| < \sqrt{2}\, d(E, F)$.

Proof sketch. The native cell Poincaré bound $\|(I - P_k)u\|^2 \le 624\, 4^{-k} \langle u, B_l u\rangle$ gives uniform compactness of the Green operators and norm convergence. Mosco convergence passes their local Markov forms to the limit; min–max estimates give the displayed spectrum. Conditional finite-population limits apply to the actual identity-or-swap repairs. The Gaussian heat bound gives finite propagation of the admitted cosine family [99, 100, 101, 89, 90]. □

Thus one source operator controls static covariance, relaxation and a chosen coherent wave service. Its spectral dimension is two. The $\lambda_n \asymp n$ count excludes identifying it, up to scale and bounded mass shift, with a three-dimensional elliptic bulk operator or a first-order boundary operator. A physical bulk interpretation needs an additional radial observation map. Passive repair supplies no wave sector; its attempt count and the coherent wave parameter require clock calibration. The native stationary reference supplies no physical quantum vacuum or universal area-entropy coefficient [72, 91]. On the binary repair sector its complete instrument has unique stationary state $I_{66}/66$.

A separate zero-commit stacked-ring control supplies vertical seams, with $L_3(k) = L_{\mathrm{ring}}(k_1, k_2) + 2(1 - \cos k_3) I_{24}$. Its acoustic Hessian has rank three and squared speeds $(15/352, 15/352, 1)$ in declared lattice units. A positive-acoustic-band packet at zero commit rate supplies position distributions and a centroid in this anisotropic flat metric. The connectivity, preparation and metric are inputs. Its centroid law is a translation-invariant band identity, not a discriminating metric or gravity test. These are finite-grid packet values, not observables of the interacting triangle source; a two-dimensional force kernel has no automatic three-dimensional transfer (`code/openwave_candidate/validation/observables/spacetime/`).

A specified radial interval and field service give the product covariance $(-\partial_r^2 + A \oplus 0)^{-1}$. If its hyperbolic coefficient density is $C = \mu\, \mathrm{diag}(-1, a)$, the effective metric is

$$g = \sqrt{-\det C}\, C^{-1} = \mu \sqrt{\det a}\, \mathrm{diag}(-1, a^{-1}).$$

The spatial coefficients are read from the source; the radial architecture, hyperbolic service and preparation are explicit design inputs.

Proposition 7.4 (Smooth effective field descriptions)**.** *Smoothing the native angular coefficients while retaining the radial Dirichlet interval gives norm-resolvent convergence, with gap* $(\pi/L)^2$*. For every fixed finite admissible menu of observer readouts on bounded wave times and every* $\varepsilon > 0$*, a smooth static* $(1 + 3)$*-dimensional Lorentzian metric carries a free scalar field whose complete predictions agree within* ε*, in addition to the stated source and clock errors.*

The proof transports the changing measures into one Hilbert space and combines compact Mosco convergence with continuity of the covariance, response and finite-history operators [72]. It proves a common finite-menu approximation, without selecting a radial architecture, providing a smoothing rate, or producing a responsive metric. The three-dimensional fermion limits in Section 9 have their own source and refinement hypotheses.

7.2 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\mathrm{Conf}^+(S^2) \;\cong\; \mathrm{PSL}(2, \mathbb{C}) \;\cong\; \mathrm{SO}^+(3, 1), \qquad H^3 \;\cong\; \mathrm{SO}^+(3, 1)/\mathrm{SO}(3), \tag{7}$$

which identify the three-dimensional space of future unit timelike directions [58]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [48]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1]. Twelve declared null-direction tests certify the conformal Lorentz class of a supplied invertible chart map. The actual single-queue controller instead has a chain of full storage dependencies; identifying physical events and matching their clock and cone to the field are separate source obligations [9, 10].

Theorem 7.5 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t, x) = t^2 - g_{\mathrm{src}}(x, x)$.*

(F1) W_{src} has dimension $1 + 3 = 4$ and Q has one positive and three negative directions;

(F2) source-unit directions, S^2 and future-null rays are exactly equivalent;

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ sends every generated precedence into the future cone whenever each parent edge obeys $g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 (h_{\mathcal{L}}(f) - h_{\mathcal{L}}(e))^2$;

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ is spacelike, then $e \preceq_{\mathcal{L}} f$ holds exactly when $F(f) - F(e)$ is future causal, so F is an exact order embedding preserving every causal interval.

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n + 1$, n the number of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 10]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [39, 40, 41].

Proposition 7.6 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [46, 47]; the local model is* $(1+3)$*, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [44, 45], an explicit tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with* G_{ab}*, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is* $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$*.*

A source-record population realizes the flat case.

Theorem 7.7 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let* $\varphi = (1+\sqrt{5})/2$*,* $L = 2/\sqrt{\varphi+2}$ *and* $q = F_n$ *the* n*-th Fibonacci number,* $n \geq 4$*. The population* $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0,L]^3$ *consists of* q^3 *distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within* $a_q = L/\sqrt{q}$*, with model duration* $\Delta_q = a_q/c$ *for a declared speed* c*. As* $q \to \infty$ *the order converges away from null separation to the speed-*c *causal order, and with* $\rho_q = q^3/(L^3\Delta_q)$*, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q *counts strict causally ordered pairs; for vertical duration* T*,* $\mathrm{Vol}(I) = \pi c^3 T^4/24$*.*

Corollary 7.8 (Proper-time ratios from record counts)**.** *If* I_q, J_q *approach interior timelike diamonds of proper durations* $\tau_I, \tau_J > 0$*, the observer readout* $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$*. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at $q = 13, 21, 34, 55, 89$ [42, 43]; the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [10].

The executable carrier has its own finite causal bound. For a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, and retained signal witnesses attain the bound; this is a speed in seams per native event. Remote repairs can carry a difference to a receiver before its first local tick, so the bound is not one seam per receiver tick [92, 73]. Complete endpoint records give a forward-cone placement in the registered response geometry, and a four-event native execution has a Lean instance proving that implication and the failure of its converse [1].

8 Gauge structure, particle operators and the pixel scale

The finite theorem fixes a current algebra under explicit response premises. A chosen carrier, memory law and field service then realize its currents and mass operators. These are different steps of one construction: the Lie type constrains admissible interactions, while the full executable specification fixes their coefficients. Recovering or choosing that specification is allowed; its agreement with physical particle records is an additional requirement.

8.1 The finite gauge theorem and matter representation

Let A be the icosahedral adjacency matrix and J its antipodal involution. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \qquad P_{12} = \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'. \tag{8}$$

The even port coordinates carry $\mathbf{1} \oplus \mathbf{5}$; the odd coordinates carry the two triplets. The response premise of Section 2.2 requires an injective derivative $D : P_{12} \to \mathfrak{u}(H)$, a commutator-closed image with positive trace pairing, and internal implementation of every proper carrier rotation. Internal implementation means that its projective implementer factors into the connected response group and a pointwise centralizer. This is the response-endogeneity clause identified as A2 in the library; it is not implied by the port count alone [103, 12].

Theorem 8.1 (Gauge Lie-type forcing)**.**
Under these complete-response and internal-implementation premises,

$$\mathfrak{g} = D(P_{12}) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

Its center is the uniform port line. The three-dimensional simple ideal carries one triplet, and the eight-dimensional ideal carries the other triplet and the five-dimensional band.

Proof. The trace pairing makes the twelve-dimensional image compact. Internal implementation makes the A_5 action inner, so it fixes the center pointwise. The transitive port action has a one-dimensional fixed space, hence the center has dimension at most one. With zero center, the only compact semisimple type of dimension twelve is $\mathfrak{su}(2)^4$. Inner automorphisms preserve its simple ideals, each of which has either zero or three A_5-fixed directions, contradicting the one-dimensional fixed space. The center therefore has dimension one; the unique compact semisimple type of dimension eleven is $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ [69]. □

The premises have an explicit finite matrix realization. Six antipodal axis vectors form a matrix U over $\mathbb{Q}(\sqrt{5})$; $\sigma(U)$ is its Galois conjugate. For even coordinates b, odd coordinates d, and $c = \frac{1}{6}\sum_i b_i$, put $\Phi_0(b - c\mathbf{1}) = \sum_i (b_i - c) u_i u_i^{\mathsf{T}}$. For nonzero real band coefficients,

$$K(f) = \operatorname{diag}\Big(\lambda_3 \widehat{Ud} + i[\lambda_1 c I_3 + \lambda_5 \Phi_0(b - c\mathbf{1})], \lambda_{3'} \widehat{\sigma(U)d}\Big), \qquad \widehat{x}y = x \times y. \tag{9}$$

Proposition 8.2 (Finite response realization)**.** *The map K is injective and equivariant, its image is commutator-closed, and proper carrier rotations act internally. Its pulled-back bracket has*

$$(P_{12}, [\ ,\ ]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3). \tag{10}$$

The six axis projectors span the symmetric matrices, while the triplets span the skew matrices in the two blocks. These rank and equivariance identities are exact over the full sixty-element rotation group. The representation and nonzero coefficients are design inputs. A zero bracket on the same port module violates the internal-implementation premise; it is not excluded by geometry alone. The gauge specialist gives the complete finite certificate and the boundary between analytic classification and machine-checked algebra [12].

Theorem 8.3 (Finite matter image)**.** *For the supplied trace-balanced module* $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$,

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L, \tag{11}$$

where $Q = (3,2)_{1/6}$, $u^c = (\bar{3},1)_{-2/3}$, $e^c = (1,1)_1$, $d^c = (\bar{3},1)_{1/3}$ *and* $L = (1,2)_{-1/2}$. *All perturbative gauge and mixed gravitational anomalies vanish, the number of weak doublets is even, and the faithful image is*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{12}$$

Among the 1,024 *subsets of the ten nontrivial irreducible exterior summands, nonemptiness, chirality and the four anomaly-trace conditions select exactly this fifteen-state mask and its charge conjugate.*

The exterior branching, anomaly traces and center kernel are finite calculations [12]. The supplied module is the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching; this result does not require a physical SU(5) gauge field. It fixes neither three families nor their mass matrices.

The source realizes this global form. The record law is covariant under independent phases on the twelve ports; read on the protected code, twelve relative port phases per seam generate exactly $\mathrm{S}(\mathrm{U}(3) \times \mathrm{U}(2))$, the group (12). Its centre is the circle $\{\mathrm{diag}(e^{2it} I_3, e^{-3it} I_2)\}$; the $\mathbb{Z}_6$ of (12) is the kernel of the covering product, and the order-six central subgroup generated by $e^{2\pi i Y}$ is the flux alphabet of the protected defects (Section 8.6). The seam cost of a code state carried around a plaquette, averaged over the code, is $1 - \mathrm{Re}\,\mathrm{tr}_5\, U_P/5$, which gives equal bare couplings at the cell and $\sin^2\theta_W = 3/8$ there; the gauge waves travel on one cone with two polarizations. The native transports alone form a finite group, and the continuous port phases are the extension that makes the links continuous. Every operation of the source conserves $B - L = N - \frac{2}{3}(N_{c1} + N_{c2} + N_{c3})$, so the source admits no Majorana pairing and its neutrinos are Dirac [12, 13].

8.2 The action and its coefficient realization

The configured ninety-mode fermion field intertwines all twelve source currents on full Fock space. Its Weyl matrices and sixty-state quantum links satisfy exact finite gauge identities. The sine regulator has eight valleys; its mirror multiplets differ from hole multiplets, and single-valley projection fails local gauge covariance. Three family copies, a Higgs doublet and canonical kinetics give tree coefficients $m_f = v\, s_f(Y_f)/\sqrt{2}$. Complete weak feedback converges to the specified mean family matrices in its pulse clock. A finite quantum Higgs extension preserves local Gauss identities with the same feedback. Vacuum selection, an interacting chiral limit and physical mass matching require separate control [13].

A prescribed coefficient matrix can also be implemented by a complete record-controlled memory process. Such an inverse construction establishes realizability of its target coefficients. It does not determine those coefficients or their physical interpretation. The particle paper gives the complete controller maps, disabled records and parameter ancestry [13].

8.3 Charged spectra of the source and the mass comparison

The charged comparison uses the native generation register and retains its complete preparation, evolution and readout. Its rest-rate coefficients have a free Dirac realization with all regulator valleys retained. The physical comparison distinguishes a derived coefficient of a declared architecture, an empirical energy anchor and an interacting mass pole [13, 73].

The charged shape has an exact identity behind it. On each oriented face the regular C_3 action has the circulant commutant $C = aI + bR + \bar{b}R^2$.

Theorem 8.4 (Positive-chamber Koide identity)**.** *If $a > 0$, the eigenvalues $\lambda_k = a + 2|b|\cos(\theta + 2\pi k/3)$ are nonnegative and $\sqrt{m_k} \propto \lambda_k$, then*

$$Q = \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\Big(\frac{|b|}{a}\Big)^2,$$

so $Q = 2/3$ exactly when $|b|/a = 1/\sqrt{2}$, and the phase of b drops out.

Equal tracial event ranks and the specified square-root lift force finite balance. Applying it to canonical Yukawa masses requires $C \propto (Y_e^\dagger Y_e)^{1/4}$; all four current-gated memory duties fail this condition. A distinct coherent program with native exclusive weights $(10, 5, 5, 2, 50)/72$ produces $Y = C^2/3$. Its full-Fock vacuum compression retains every ancillary quantum outcome. The same coefficient has a controlled free Dirac limit with all lattice valleys retained; interacting chiral propagation and physical clock matching require separate proofs [13, 14].

The balance and the phase have a source. In the coherent repair law the repair event is a swap of the seam's port qubit with its record, so the record holds the source endpoint in its conjugate basis, and the oriented record keeps the repair branch. Two seam objects C and E on one face, a neutral register carrying the face third-turn R, and the clock $K_\theta = e^{i\theta J}$ on the oriented current J of C form one tick $U(\theta)$.

Theorem 8.5 (Generation register)**.** *The 432-state tick is unitary, and its compression on the Fourier analyzer of the three flags "E agrees", "C repairs +", "C repairs −" is $M(\theta) = \frac{\sqrt{2}}{3}C(\theta)$ with $a = 1/\sqrt{6}$ and $|b| = 1/\sqrt{12}$. Hence $|b|/a = 1/\sqrt{2}$ and $Q = 2/3$ for every phase in the positive chamber.*

The balance is the count of branch weights $1/6 : 1/12 : 1/12$ of the agreeing hold and the two oriented repairs. The phase is a count as well: the register's exclusive read has probability $1/12 + 1/6 - 1/36 = 2/9$. On the energy clock, where $\langle I - \mathrm{Re}\, U\rangle = P(\text{record } 1)$ holds on every blank-record input with uniform address, N weak rounds give the oriented current the phase $2/9$ with error at most $1/N$. The clock fixes its own unit: since $I - S$ has eigenvalues 0 and 2, its edge term $(\pi/4)(I-S)$ run for clock time τ reproduces the tick's root-swap gate exactly when $\tau \in 1+4\mathbb{Z}$, and the principal value is one unit of clock time per tick. With masses proportional to λ_k^2,

$$\frac{m_\mu}{m_e} = 206.770316, \qquad \frac{m_\tau}{m_e} = 3477.4728,$$

against the frozen CODATA 2022 ratio 206.7682827(46) and the PDG 2024 tau average. The muon discrepancy is +9.83 parts per million; the electron-anchored tau energy differs by +30.94 parts per million, or approximately 0.61 quoted standard uncertainties. These marginal comparisons supply no joint likelihood. The architecture was developed with particle masses and Koide relations known, so this is a retrospective conditional comparison. No radiative correction is applied to remove the muon discrepancy [119, 25].

The analyzer weights, oriented record, exclusive read and unit kick coefficient are construction inputs. Native event probabilities fix the count under those preparations and reads; they do not select its conversion to a phase or the squared-eigenvalue mass interpretation. These choices retain their target-informed ancestry even when their complete execution is verified.

The count survives many ticks. One weak round is the tick, the kick on the oriented current, the inverse tick, a return test and a reset; every round reads the first-tick weights, and N rounds per unit clock time give the phase $2/9 - 35/(2187N^2)$. Executed on the sixty-address carrier for up to 2×10^6 rounds, the phase follows this form and tends to $2/9$ within 10^{-13}, with every outcome

of the return tests kept. The three characters of the face third-turn are the eigenvectors of the executed mass round, with rest rates $y_k = \frac{1}{27}(1+\sqrt{2}\cos(\frac{2}{9}+\frac{2\pi k}{3}))^2$. With the electron mass as the dimensionful calibration, these conditional coefficients give

$$m_\mu = 105.659414 \text{ MeV}, \qquad m_\tau = 1776.985 \text{ MeV},$$

against 105.6583755(23) and 1776.93 ± 0.09 MeV. The register and analyzer are a construction on the coherent law. The operation that brings the pair onto one port, the source event of the kick require their stated apparatus implementations; the electron mass supplies the absolute energy anchor. This comparison does not identify interacting physical lepton poles [13, 119].

A finite-range Higgs–Wilson extension gives exactly one charged minimum per family. With $x_j = ap_j$ and positive analyzer mass m, its complete charged dispersion obeys

$$E^2 = \frac{\sum_j \sin^2 x_j + (am + \sum_j (1-\cos x_j))^2}{a^2}, \qquad \partial_{x_j} E^2 = \frac{2\sin x_j}{a^2}\left[am + 1 + \sum_{k\neq j}(1-\cos x_k)\right].$$

The derivative is positive for $0 < x_j < \pi$, so $p = 0$ is the only charged local minimum. Native charge and colour Casimirs identify exactly three colourless charged Dirac families. Every other corner mode and all neutral zero modes are retained. The resulting masses in GeV are

$$(m_e, m_\mu, m_\tau) = (0.00051099895069,\ 0.1056594144959,\ 1.7769849708134).$$

The declared Wilson coefficient is one and the rest values are unchanged from the analyzer. For momenta of norm at most P, the full symbol differs from its continuum Dirac comparator by at most $aP^2/2 + a^2P^3/6$. Contractivity gives the corresponding complete-record error at finite time. This proves a finite charged-family count, without transferring a chiral gauge-continuum theorem or a physical interacting pole identification (`model/openwave_single_valley_leptons`).

8.4 A common positive source for lepton and hadron records

A positive two-cell native action uses colour, spin, weak and baryon-density currents, together with graph hopping. Four empirical targets select its coefficients: the proton, neutron and charged-pion energies and the axial magnitude. Complete colour-singlet sectors retain 120 baryon and 64 meson states. The actual weak vector coefficient is exactly one, and an integer-state residual certificate gives

$$g_A \in [1.2752999999554, 1.2753000000445].$$

No multiplicative correction is applied to the native current. The state has approximately 96.42 percent split-site probability, outside the co-located multiplicity-one sector whose axial coefficient is 5/3. The fitted energy records in the unchanged electron unit are

$$(E_p, E_n, E_{\pi^+}) = (0.93827208943, 0.93956542194, 0.13957039) \text{ GeV}.$$

These three energies and g_A are calibration inputs, not independent mass or current predictions [120].

The same strong action has an explicit extension with 192 native fermion modes: all 144 colourful and 48 uncoloured coordinates, three charged analyzer coefficients and three calibrated neutral Dirac coefficients. At zero spatial lepton kinetic coefficient, couple its actual quark and left-lepton currents by GV, with $G = 1/64$. The complete local one-quark/one-family current has norm 6, hence

$$|V| \le 18N_Q, \qquad H \ge (m_{\min} - 18|G|)N_{\text{colourful}} + H_L, \qquad m_{\min} - 18|G| > 1.$$

The quark vacuum is reducing and the massive lepton reference is unique. Consequently the full charge ± 1 threshold is exactly the electron anchor 0.00051099895069 GeV. All six Dirac branches coexist on this positive matter source. Photons and purely leptonic contacts are absent from this action; no gap statement is transferred to a photon-containing theory [121].

The pure Dirac reference has active covariance $I/2$ and zero active–active anomalous covariance. Wick contraction gives $\langle V \rangle = 0$ and $\langle V^2 \rangle = 3 \sum_r \langle J_{Q,r}^2 \rangle$. For the actual native integer baryon trial, its full energy variance is $r^2 + G^2\kappa$, where r^2 is the exact strong residual variance and $\kappa \simeq 36.091699949922$. The spectral theorem then gives at least 0.964754199267 conditional probability in a window of half-width 0.255499475345 MeV around either calibrated nucleon mean. The probability from the original occupation seed exceeds 0.042440240225; failure and the outside-window result are retained. This exact Gaussian reduction avoids a 2^{48} lepton-Fock enumeration without truncating the source. The windows are finite spectral records, with no identification of physical poles or resonance widths.

A positive hyperfine member adds the native spin gradient $\lambda \sum_a (S_0^a - S_1^a)^2$, with $\lambda = 5/2$, and the positive quark–antiquark pair interaction

$$h \left[\frac{3}{2} N_q N_{\bar{q}} + 2 \mathbf{S}_q \cdot \mathbf{S}_{\bar{q}} \right] = h \sum_{i \in q, j \in \bar{q}} (\mathbf{s}_i + \mathbf{s}_j)^2.$$

Seven disclosed inputs select its proton, neutron, charged-pion, axial, deuteron-binding, neutral-vector and Delta-assigned agreements. One unchanged electron unit gives

$$\begin{aligned} (E_p, E_n, E_{\pi^+}) &\simeq (0.93827208943, 0.93956542194, 0.13957039) \text{ GeV}, \\ (E_{\rho^0}, E_{\Delta^+}, E_d) &\simeq (0.77526, 1.232, 1.87561294497) \text{ GeV}. \end{aligned}$$

The vector and Delta targets are real Breit–Wigner mass conventions, without a derived complex resonance pole or width. All 120, 64 and 2520 baryon, meson and nuclear singlets are retained. Comparison with every compatible two-baryon threshold gives no additional bound nuclear channel. The neutral spin-zero candidate is 0.13827705749 GeV, which misses its unused neutral-pion comparison by 3.30025749 MeV (`model/openwave_common_hyperfine_spectrum`).

Exact local SU(3) fusion gives $\sum_x C_x^2 \le (22/7) \sum_e E_e^2$ on this matter/Gauss carrier. The resulting full-source positivity permits a shared length $a \simeq 0.75590872875$ fm, selected by the quadrupole and illustrative 1 GeV/fm tension. Pinning a quark and diquark at nonadjacent interior sites leaves all 40 internal states and gives the exact total-source identity $H_{\mathrm{pin}}(r+1) - H_{\mathrm{pin}}(r) = (4\kappa/3)I$. The unconstrained two- through five-cell calculations retain complete spectra and their native currents. Their axial response changes from the fitted 1.2753 to about 1.66549, so increasing graph size does not preserve that fit. Exact Schur elimination supplies the full complement memory, normalization metric and current vertices (`model/openwave_strong_step_scaling`).

Calibrated interacting source poles. The complete native three-quark Gauss source has five compact cluster fields per translation cell, with matrix spectral measure of zeroth moment I_5. Exact bounds retain every relative configuration outside this frame; the squared compact-to-complement coupling norm is 5/64. Two declared counterterms vanish on the entire calibrated two-cell operator. With the inherited neutron and axial inputs, the neutron rest pole lies in $[0.939565387072, 0.939565456806]$ GeV, and the native axial magnitude in $[1.274757, 1.275843]$; native isospin gives the proton shift and vector amplitude one. The fixed-coefficient control has disjoint axial interval $[1.291676, 1.292715]$, so preserving two-cell fits alone does not preserve spatial refinement.

The ground pole is simple at every momentum in the Brillouin circle. Splitting the five compact coordinates into three light and two high coordinates, then removing the common scalar propagation, gives a uniform isolation gap above 0.127 source units and complete high-sector probability below 2.7×10^{-6}. The underlying positive correlated kinetic rule remains $J=\zeta M^{-1}$, with inherited electron unit, length and speed. Every spectral complement is retained (`model/openwave_uniform_pole_bands`).

The explicit wave amplitude

$$f(k)=\sqrt{\frac{15}{16\kappa}}\left[1-(k/\kappa)^2\right]\mathbf{1}_{|k|\le\kappa},\qquad \kappa=1/512,$$

is normalizable on the infinite chain and has finite position variance. Uniform projector bounds give neutron and proton band probabilities above 0.9999614; conditional on that outcome, more than 98.4025% lies within 8192 cells. Complete native-current packet responses retain preparation, band, current and position failures, reference charge and work. The exact band filter is a declared nonlocal spectral service, without a finite local compiler or physical four-dimensional particle identification (`model/openwave_normalizable_particle_records`).

Native strange particles and their breakup spectrum. Two coefficients calibrate the complete two-cell kaon and Lambda energies to 0.493677 and 1.115683 GeV; unused neutral-kaon, Sigma and Xi comparisons fail. The calibrated bulk Lambda rest pole lies in [1.115684628607, 1.115684878081] GeV. A normalizable native packet with $\kappa=1/512$ has leftmost-cell variance 655360 and full ground-band probability above 0.9997. All 72 native weak CAR words preserve that spatial packet; exact internal Gram contractions retain every current output without requiring a band filter (`model/openwave_lambda_wavepacket`).

The full five-body strong Hamiltonian has a Weyl sequence at the Lambda rest energy. The literal 72-word current instead has total successful branch mean $[2.418046697462, 2.418046697463]$ GeV and variance $[0.085780468809, 0.085780468810]$ GeV^2. Its roughly 1.302 GeV conditioned increase is distinct from nonselective work. For the complete spectral projector $P_w=1_{[E-w,E+w]}(H)$,

$$\|P_w u\|\le\|u-(H-E)x\|+w\|x\|.$$

Rational witnesses with eight displaced proton–pion cluster trials retain all translation interference and Hamiltonian successors. They bound the conditional weight within 1 MeV of the rest energy below 0.026622. The positive narrow-window lower bound and outgoing spectral density are unproved; neither spectral inclusion nor this upper bound gives a lifetime (`model/openwave_lambda_narrow_weight`).

Nuclear isospin and native charge response. An occupancy-gated three-cell spin interaction vanishes on every sector with at most six quarks, preserving all seven two-cell calibration domains. One triton mass input fixes its coefficient. Exact colour exterior characters reduce complete 572572- and 176729280-singlet carriers to 64 and 256 low coordinates with full complement bounds. The triton-assigned interval $[2.8089192030, 2.8089230704]$ GeV contains its 2.80892113668 GeV input (`model/openwave_light_nuclear_curve`).

The native isospin intertwiner gives a triton–helion separation 1.29333251 MeV against the supplied 0.52952556 MeV comparison; charge-breaking dynamics is required. A complete compact $U(1)$ field leaves held-out helion interval $[2.807625870510, 2.808017057294]$ GeV. A neutral harmonic field coupled to a four-site occupation projector O_b provides a stable signed extension:

$$H\ge H_{\mathrm{EM}}-\frac{g^2}{2\omega}\sum_b O_b.$$

With declared $g = 1/8$, $\omega = 1/16$, it vanishes below twelve quarks and retains every oscillator level. Exact coherent-state trials enclose alpha in $[3.739060107335, 3.739677967934]$ GeV. Both held-out masses fail; no target selected these coefficients. A full-singlet Casimir trace gives unconditioned electric preparation energies 8955/9152 and 2949/1360 source units on three and four cells, retaining all charge sectors (`model/openwave_nuclear_field_attraction`).

The literal native isospin current transports the neutron ground ray to the proton and gives the same 1.29333251 MeV mass ordering. The neutron has positive radial charge at graph radius 1/3, negative charge at 2/3 and zero charge at the origin. Complete compressed occupation instruments retain the 1/120 preparation probability and all failures. With the shared length, the proton radius is about 0.33456816 fm and the neutron charge second moment about -0.03731194 fm^2. Their ratio excludes the experimental comparison box for every common length. The qualitative profile match therefore coexists with a failed quantitative radius test; no relativistic Sachs form-factor identification is assumed (`model/openwave_baryon_form_factors`).

The five-input spin-gradient control omits the pair-triplet fit and has an unassigned spin-three bound nuclear channel. A common total-spin-square control cannot fit both rho and Delta. These statements concern those specified actions, separately from the complete seven-input spectrum.

Native weak response on the same source poles. Exact contraction gives vector matrix I_5 and compact axial matrix $5/3 \oplus \left(\begin{smallmatrix} 1 & 2/3 \\ 2/3 & 1 \end{smallmatrix}\right)^{\oplus 2}$. The complete fixed-weight axial square M obeys $(M - I)(M - 4I) = 0$, retaining all spin and isospin outputs outside the pole frame. Exact complementary and trial-vector bounds transfer the native currents to the interacting poles at six momenta: the rest vector amplitude is one, the calibrated rest axial interval is $[1.274757331914, 1.275842668086]$, and the vector amplitude at phase π exceeds 0.994488237318. A charge-conserving current instrument retains success, failure, complementary spectral outcomes and bounded source work. At rest its vector success probability is 1/9, with 8/9 failure and mean source work $-1.29333251/9$ MeV. These are complete source-current records, without an autonomous weak decay or physical lifetime (`model/openwave_pole_weak_current`).

An open-chain colour source admits an exact attachment to this two-cell strong action. On every global colour singlet, $C_0^2 = C_1^2$, so replacing U by $U - \kappa/2$ and the vanishing global colour penalty by κC_0^2 preserves the whole two-cell singlet operator. Native Gauss elimination on a longer chain gives

$$E_x = \sum_{y \le x} C_y, \qquad V_{\text{string}} = \kappa \sum_x E_x^2 = \frac{4\kappa}{3}(x_{\max} - x_{\min})$$

for the complete three-quark singlet. Positive terms give a uniform bound on its mean relative span at every finite chain length. All 120, 364, 816 and 1540 singlet states are retained on two through five cells. No coefficient is refitted: the last two neutron-assigned ground energies are near 0.939664299718 GeV, while their axial response differs from the two-cell fit. A physical length is a separate attachment. If the same length also matches the supplied deuteron quadrupole benchmark, the $\kappa = 4$ tension is about 0.00360537 GeV/fm. The sufficient restriction $\kappa \le 2U$, which makes each local colour coefficient nonnegative, gives tension below 0.428678 GeV/fm. Full-operator positivity has a larger domain: an exact local Fock decomposition gives $C_x^2 \le 3N_x$, and Gauss incidence bounds $\sum_x C_x^2 \le 4 \sum_e E_e^2$. The positive number energy controls the negative local coefficient. This permits the common quadrupole/string attachment near 0.755908729 fm per cell and 1 GeV/fm on the five-input source. Both dimensional comparison values select that attachment; neither is an independent prediction (`model/openwave_string_positive_cone`).

On the distinct four-input axial/common-Dirac member, a longitudinal electric extension dresses every charged hop and cross-cell weak product with compact rotor shifts and retains the com-

pensating quantum charge reference. Its complete Gauss kernel is isometric to the entire 192-mode matter space. A declared graph-Green electric energy and its fixed diagonal self-energy subtraction preserve the one-lepton rest values and the exact electron charge threshold. The baryon trial means shift by about -1.629 and -6.298 keV for the neutron and proton assignments; their full-source spectral windows are recomputed without a mass refit. The action contains longitudinal flux coordinates, with no transverse photon or fine-structure-constant identification (`model/openwave_common_electromagnetism`).

A compact-field source uses the native charged spin coordinates and $\sqrt{m^2 + (\boldsymbol{\sigma} \cdot \boldsymbol{\Pi})^2} - m$, including its Pauli magnetic commutator. All integer fluxes, both spin branches and the compensating reference remain in the controlled source. A positive Wilson term lifts all seven extra free corners, and the free matter and quadratic Maxwell symbols share $c^2 = 1/128$ under controlled low-momentum refinement. This intraband source preserves excitation number; it supplies no pair-producing vacuum or interacting Lorentz theorem (`model/openwave_dirac_maxwell_refinement`).

A local record instrument with exact compact persistence. A specified one-load realization uses ports and transit seams of the canonical ring screen, with two six-level half-seam records per seam. Its lifted generator has $H_m = D + m\Gamma$, where an endpoint-to-transit hop increments its own record by $+1$ at either endpoint. Its Hamiltonian amplitude is $+1$ at the first endpoint and -1 at the second; Γ is $+1$ on ports and -1 on transit seams. It preserves Gauss charges modulo six. Integer records with a hard wall define a different regulator; no integer-charge theorem follows from identifying residues.

For a connected screen with N ports and M seams, each fixed-position Gauss fiber contains $d = 6^{M-N+1}$ record configurations. Let $V|z\rangle$ be their normalized uniform superposition. Half-hop translations and every local C/E outcome map are bijections between the relevant fibers, so

$$H_{\mathrm{lift}} V = V H_{\mathrm{bare}}, \qquad K_\alpha^{\mathrm{lift}} V = V K_\alpha^{\mathrm{bare}}.$$

Thus the reduced program computes every local-record history and position readout, including arbitrary idle quantum references, on this explicit preparation domain. A definite record preparation does not satisfy the same intertwining relation. The C outcome groups sources by coin and current; E groups them by coin and contrast. This local-coarse quantum extension is supplied. Complete source/record monitoring defines another instrument.

The canonical carrier face gives a normalized signed three-seam cycle χ with $B\chi = 0$. The transit state $\Phi = V(0, \chi)$ obeys

$$H_{\mathrm{lift}} \Phi = -m\Phi, \qquad K_{e,\ell,c,0} \Phi = \tfrac{1}{2}\Phi, \qquad K_{e,\ell,c,r\neq 0} \Phi = 0.$$

Each of the four zero-record outcomes of a drawn seam has probability $1/4$. Its density therefore persists under the same coherent evolution, local commit and supplied per-seam continuous scheduling. The three supported transit seams each have probability $1/3$; all port probabilities vanish. This conditional stability uses the declared transit-as-elsewhere rule.

The uniform state is compact in position and spread over $3d$ joint basis states: $d = 6^{40}$ on the two-carrier screen and 6^{157} on the eight-carrier screen. Native holonomy-preserving maps cannot raise a definite record's initial code weight $1/d$. Added modular Fourier gates and reversible tree Gauss arithmetic prepare the uniform fibers, including composite modulus six. These are explicit resources. Individual record measurements retain their full complementary poststates for continuation; whole-position closed-cut measurements preserve the reduced code. The compact mode is flat and immobile, with no internal clock or identified electron, magnetic moment or physical angular momentum.

The uniform-fiber construction also supplies charge-only pinned-source spectroscopy, complete negative branches and the current $i[H, P]$. Its standing eigenstates lose stationarity under native monitoring. Complete energy/event apparatus retains reference conditioning and supported battery work; its spectral access and 511 keV half-gap calibration are supplied inputs, with no hydrogen or physical species identification (`code/openwave_candidate/candidate/methods/pinned_charge.md`).

A separate literal native construction joins spatial integer charge to an actual native apparatus without a coherent-control service. Its source retains the initial occupation G, has total load one or two, and starts with zero current tape F. On the carrier's punctured dual sphere, a local circle-valued realization of the emitted current has regional degree $\sum_{v\in A}(BF)_v = \sum_{v\in A}(G_v - n_v)$. Exact integral incidence and face-boundary minors establish $\ker B = \operatorname{im}\partial_2$. The local spatial map and reference are declared choices; the abstract lattice does not select a unique physical interpretation. Integer degrees add across disjoint daughter contours, so a primitive unit cannot split into two equal half-charge outcomes. A native double load can split into two whole units.

A connected thirty-event contrast sweep checks the current reconstruction against the actual source, conditional on true G and faithfully received closing records. The sweep disturbs the source; the charge refers to its final latch. Modulo-six counters have a unique bounded lift on the regional range $[-2, 2]$. An initially unknown one-load carrier resets through native currents and records the signed degree by movement around a six-edge cycle. The complete tape is validated before decoding, with no supplied terminal-state snapshot in admission. All discovery, routing and measurement timeouts retain their probability and report no charge. The portable experiment checks all 5400 native branches, 6228 preparation/state pairs and 172 complete signed or zero readouts, with 768 complete probability laws supplied by an independently checked exact recursion (`code/openwave_charge_quantization/PROOF.md`). The result establishes static relative topological charge on this specified construction; it supplies no electron mass, unrestricted modulo-six decoder or transfer to the interacting fermion–field source.

A declared CAR extension retains direct fermionic endpoint phases in every native C/E outcome. The interacting triangle instance contains all 64 occupations of six orbitals and six modular Gauss fibers per occupation: 384 source states. The integer boundary record $B = 3 - N_{\mathrm{source}}$ dresses a complete two-occupation reservoir, giving 768 joint coordinates. One Hamiltonian joins CAR half-hopping to a periodic half-record potential and a cyclic face term, with supplied $m = 0.23$, $\kappa = 0.37$, $g = 0.61$. Its recomputed all-number ground is unique in $N = 3$, at -2.7262233938630343. This is the selected finite reference; bare character-sector spectra are zero-field controls, not imported gaps.

Every dressed reservoir addition, removal and no-change outcome retains its initial reservoir label, source-number/energy records and full poststate. Source charge plus reservoir charge is conserved. The pulse commutes with the record-field energy, so its unconditioned field energy is unchanged although conditional branches differ. Work uses this same interacting $H + 0.23 n_{\mathrm{reservoir}}$ with explicitly zero extra boundary energy. The selected native protocol evolves for 0.37 hop units and commits MIX with probability 0.29, retaining the 0.71 no-commit outcome; its mean monitoring work is 0.5161569571 source units.

On this same source, a declared external Peierls phase couples to the half-hop current $J_h = -\partial H/\partial A_h$. Complete native records, state preparation and readout transform together under the orbital gauge change. This is a controlled external intervention, not a dynamical Maxwell field or magnetic-moment measurement. All 1392 charged-energy bins retain their original weights under one display calibration: the dominant addition line is assigned 0.000511 GeV. The removal/addition peak ratio is 4.892537374; no particle identity or mass prediction follows from that supplied anchor.

Grouped number/energy measurements retain complete degenerate eigenspaces and their full

coefficient matrices for native continuation. Independent basis rotations preserve those poststates; resolving a fine basis and forgetting its label does not. A complete marked clock combines this H and the native map Φ through $\mathcal{L}(\rho) = -i[H, \rho]+\lambda(\Phi(\rho)-\rho)$. Two native regional charge instruments and all 348 energy groups compose on that clock with retained reset, export and multiple-event branches. The integer-field comparison bounds the complete record law by 4.185417×10^{-6} at modulus 42; its modulus-six bound is one. The larger-register precision does not certify the smaller particle calculation. The distinct reciprocal classical force calculation has no proved dynamics transfer to this source (`code/openwave_chain_checks/provenance/PROOF.md`). Statistics, field coefficients, spectral preparation, apparatus and the dimensionless clock are declared inputs. The triangle supplies no physical species, spin, vacuum, calibrated mass, distant radiation or gravity identification (`code/openwave_candidate/next/C10/CANONICAL_CONSTRUCTION.md`). The executable-construction paper gives the proof [72]; the reduced operator and complete records are specified in `code/openwave_candidate/candidate/methods/uniform_native.md` and `code/openwave_candidate/candidate/methods/compact_native.md`.

A separate cubic source couples 2048 padded fermion modes to one shared 288-edge, modulus-sixteen field. Its declared step combines the mass coin, current-controlled transport, a face kick and the electric phase, with $h = 1/8$, electric coefficient 0.17 and face coefficient 0.41 on one face. A two-charge preparation and inverse-transport analysis give a connected phase $e^{0.17i}$. The ideal event-law separation from the additive-field control is 0.0054057124, with native implementation allowance below 3.29647×10^{-5}. All 64 ideal joint event/field outcomes and the native instrument's additional failure records are retained. A distinct three-step, two-mobile-charge experiment changes the pair-position law by total variation 0.0004639073 when the field couplings are switched off. These are finite interaction records with specified preparations and apparatus, without a Coulomb-strength or physical-particle identification. The shared-source, clock, quantum-field and reference experiments have isolated executable replays and separately typed preparation domains (`code/openwave_common_dynamics/PROOF.md`).

A homogeneous noncompact field on the routed carrier has a controlled vacuum Maxwell limit. Its actual oriented face matrix gives the curl factorization and geometric coefficient $\rho = 2/3$. With declared electric and magnetic coefficients $a = 1/2$, $b = 3$, the two transverse branches have $c^2 = ab\rho = 1$ in the source's length and macro-time units. Native field and clock instruments retain reset and export failures. A size-three execution includes all 646 coexact modes in the vacuum-plus-one-quantum code; autonomous spectator reduction supports the separate controlled radiative limit. Speed matching is a coefficient choice, with no empirical coupling prediction.

The distinct scalar mass-coin walk has a 144-coordinate Bloch cell and a rank-26 slow subspace. Its source-derived gradient Gram matrix is $g_a^* g_b = \delta_{ab}/48$. On the prepared scalar branch, $\omega^2 = m^2 + |p|^2/48$, and native Ramsey records converge to the massive scalar return amplitude with a complete preparation, source, clock and readout bound. Flat and optical branches retain their full probability. The coordinate change $X = \sqrt{48}\, x$ writes the usual Klein–Gordon equation; it does not identify this walk with the routed Dirac source. Both wave criteria have reproducible full-instrument reviews on their specified domains (`code/openwave_wave_rows/`). Large refinement words are finite certified programs, without a claim that every pulse was literally executed. An interacting common-speed limit and physical mass spectrum require additional source consequences.

8.5 Native outgoing currents and neutral mixing

The unchanged native semileptonic current contracts with the actual calibrated baryon states. Its allowed point-contact coefficients are

$$a = -0.10654396, \qquad A = -0.11943525, \qquad B = 0.98710871, \qquad D = 0.$$

Complete four- and eight-site three-dimensional cubes retain all momentum, spin, family and hadron outputs. The full-Fock remainder bounds a strictly parity-odd momentum record, including preparation failure and all multipair complements. The calculation is also executed on the spin-gradient baryon state with its certified axial coefficient. Momentum and group velocity are distinct on parts of the sine regulator. The point-contact coefficients are a conditional consistency comparison: the supplied axial datum shares experimental ancestry with angular measurements, and a and B do not match their named precision comparisons. The finite two-contact records are not substituted for point-contact coefficients, physical continuum endpoints or a neutron lifetime (`model/openwave_beta_angular`).

For the positive-frequency leptonic source, exact native spin and three-body phase-space contractions give

$$\Gamma = \frac{G^2 M^5}{1536\pi^3} f(m/M), \qquad G = 2\sqrt{2} G_F.$$

All nine massive-neutral pairs are retained for each charged transition. The calculated muon-electron partial width is $3.00876260 \times 10^{-19}$ GeV. The tau-electron and tau-muon widths are $4.04898950 \times 10^{-13}$ and $3.93790076 \times 10^{-13}$ GeV; their ratio 0.972563836 agrees with the unused PDG 2025 comparison 0.9730 ± 0.0022. The supplied G_F has muon-lifetime ancestry, so $\hbar/\Gamma_\mu = 2.1876500\,\mu$s is a normalization check; it differs from $2.1969811\,\mu$s by -0.4247 percent without radiative corrections. Hadronic tau channels and a total tau lifetime are outside this source.

An exact coarea reduction bounds the difference of massive and massless reservoir amplitudes by $\|\sqrt{\rho_m} - \sqrt{\rho_0}\|_2 \le 2\mu_{\max}^2$ in parent units. Duhamel then bounds inclusive survival-probability error by $4Gt\mu_{\max}^2$. This transfers the finite Volterra lifetime bound to the calibrated massive-neutral member, including all angular dark channels. The below-threshold comparison adds uncoupled zero-amplitude channels and does not identify individual angular records. The weak normalization is calibrated; physical observer-clock emergence is a separate condition (`model/openwave_lepton_widths`).

For the native meson current, the exact injection identity $C^\dagger C = 24I$ reduces all nine emitted-family blocks to one operator plus scalar energy offsets. A cubic complete-source remainder gives an infinite-line light-channel probability above 7.8898788×10^{-8} at source parameter 0.1 from a 128-site calculation, including volume and numerical errors. The original-input bound is 1.8935709×10^{-8}. This proves finite emission without installing an irreversible decay law [123].

A separate 288-mode Dirac source calibrates two neutral mass splittings, three mixing angles and one phase to the specified NuFIT 6.0 data edition. Normal ordering and the lightest mass 0.001 eV are declared choices. The conditional mass eigenvalues are

$$(m_1, m_2, m_3) = (1.0000000, 8.7120606, 50.1398045) \times 10^{-12} \text{ GeV}.$$

Its unchanged native current, massive dispersion and complete active/sterile absorption instruments generate the corresponding vacuum oscillation law within total variation 4.1×10^{-9} through atmospheric phase 64π. Production failures and antiparticle channels are included. At phase π, executed muon-flavour neutral and antineutral electron-detection probabilities are approximately 0.047942 and 0.034650; their difference follows from the calibrated phase. The rigid common multiplier of the charged hierarchy fails both normal- and inverted-order splitting comparisons. No independent neutral-mass prediction or physical beam, baseline or detector calibration is inferred [124].

A direct contraction of the native forward electron-neutral current has characteristic polynomial $x^9(x-1)(x+1)^2$. Its unpolarized electron mean is $(P_e/2 - I/4) \otimes I_{\text{spin}}$, giving the matter potential $V_{\text{CC}} = \sqrt{2} G_F n_e$ with the common neutral-current term retained. Complete fresh-electron spin

instruments approach this coherent mean with an explicit channel error. In a declared 2.5 GeV, 1300 km medium, the electron-appearance probabilities are 0.0697011 and 0.0197144 for neutrino and antineutrino, compared with vacuum values 0.0491046 and 0.0351399. Both helicities, sterile coordinates and all spin records are retained. These are scenario calculations under a forward-current port architecture, not a measured beam-event fit or a transfer from unfiltered momentum-changing scattering (`model/openwave_neutrino_matter`).

These finite instruments reuse the complete-projector and reference- stability theorems after checking their source, preparation and current identities. Filled-hole, spatial Dirac and zero-kinetic Gaussian references are distinct domains. Their spectral lines and decay laws cannot be transferred by identifying their labels.

8.6 Protected centre classes and particle observables

On the separately declared gauge carrier, executable observer runs carry a protected register on its faces. A face holds a flux class $k \in \mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$, a colour-centre charge with a weak-centre parity. Accepted twists create conjugate pairs, which move, fuse and annihilate, and the classes on one closed carrier add to zero. A quantum's internal pattern is an occupation (n_3, n_2) of the five-level code $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$, with $6Y = 3n_2 - 2n_3$.

Theorem 8.6 (Native flux and centre-character dictionary)**.** *The kernel element* $(\omega I_3, -I_2, e^{2\pi i Y})$ *of the covering product acts as the identity on all* 32 *patterns, and a flux of class* f *acts on a pattern by the phase* $e^{2\pi i f \cdot 6Y/6}$*. The class seen by a pattern is*

$$k \equiv 6Y \pmod 6, \qquad k \bmod 3 = n_3 \bmod 3, \qquad k \bmod 2 = n_2 \bmod 2,$$

and the class multiplicities are $(4, 6, 6, 4, 6, 6)$.

Proof. ωI_3, $-I_2$ and $e^{2\pi i Y}$ act by ω^{n_3}, $(-1)^{n_2}$ and $(-1)^{n_2}\omega^{-n_3}$; the rest is reduction of $6Y$ modulo 3 and 2. □

Table 2: Native flux classes and compatible Standard Model centre characters. A class does not uniquely identify a physical species.

k	Centre characters	Particles	In the level-10 run
$1, 5$	$\omega^{\pm1}$, -1	left-handed quarks Q and antiquarks	mobile; $1 + 5 \to 0$
$4, 2$	$\omega^{\pm1}$, $+1$	right-handed quarks and antiquarks	frozen after sweep 15
3	1, -1	lepton doublet and Higgs doublet	frozen
0	1, $+1$	e_R, ν_R, photon, gluons, W, Z	no defect

A quantum of class k taken around a flux of class f gains the phase $e^{2\pi i f k/6}$, so colour flux is seen only by quarks and weak flux only by the weak doublets, the left-handed fermions and the Higgs. The central involution z of the carrier's rotation cover acts on the code as $\mathrm{diag}(1, 1, 1, -1, -1) = e^{i\pi \cdot 6Y}$; on the Gauss sector the 2π rotation equals $(-1)^N$ for N quanta. Of the gradings allowed by the source's vertices exactly one agrees with it: every pattern is a fermion, and defects, links and the Higgs register are bosons. The grading rests on the per-quantum spin doublet, the Gauss law and the antisymmetric occupation of the quanta. The fermions are the quanta that carry the class; the defect is the flux they see, and a baryon is three quanta on one carrier.

All 16 reaction types of the runs conserve the class. The frequent ones carry the centre classes of the Standard Model vertices: quark–antiquark annihilation $1 + 5 \to 0$ (609,494 events at level

10), the quark mass vertex $1 + 3 \to 4$ (238,523), the quark Yukawa vertex $1 + 2 \to 3$ (315,024), lepton pairs $0 \to 3 + 3$ (89,909), and the diquark channel $1 + 1 \to 2$. Carrier contents are class neutral: $\{2, 4\}$ on 69.2 percent, $\{1, 5\}$ on 14.4 percent, $\{3, 3\}$ on 10.9 percent, and the quark and antiquark triples $\{4, 4, 4\}$ and $\{2, 2, 2\}$ on 1.2 percent each. No carrier holds a lone colour class.

The runs start from a blank register and random loads. Every class appears in pairs in the first consistent slice, left-handed quark pairs then annihilate, and right-handed quark and lepton classes freeze once the loads balance. Matter and antimatter come out equal within the run's resolution: 86,476 quark triples against 86,788 antiquark triples. This is a colour-neutral, approximately conjugation-symmetric native register mixture. The runs carry no physical temperature or identified particle abundance. Their comparison with a hot-plasma census is conditional, and they supply no baryogenesis prediction.

Keeping the integer current of every twist lifts the register to integers. On the runs at levels four to seven the lift reduces to the stored class on every face of every stored cut, every carrier total is zero, and 99.7 percent of defect faces carry a value $6Y$ of a Standard Model multiplet; the lift is no conserved label of the member inside a class.

The class is the centre class. It carries neither the generation nor T_3, so up and down, electron and neutrino, and colour singlet and octet share a class; the generation sits in the register of Theorem 8.5. The executable certificates check every pattern, grading and reaction in exact arithmetic [13, 1, 3].

8.7 Localized modes and their energies

A physical massive particle requires compatible localization, energy, transport and detector records. Actual native Fock charge operators have a discrete lattice containing lepton units and quark charges $2/3$ and $-1/3$. Complete arrival instruments carry the same central current and regional Gauss flux. Finite spin-half bands and literal cube rotations retain $J^2 = 3/4$ and the odd-state 4π return relative to an idle reference. These results identify exact finite current and representation structures; a texture winding is a distinct invariant [125].

The full spatial/photon charged source has a controlled spin-half band in its stated trap domain and shell convergence. Trap removal gives a threshold with no normalizable minimizing state, so it supplies no self-bound electron. The complete infinite-photon current has static coefficient in $[1.99827530667, 1.99827540121]$, with finite-time precession bounds. Its proximity to two is a spin-half response comparison; the deviation does not reproduce the measured electron anomaly. Field normalization, trap and regulator are specified inputs [125].

The native current also converts a particle–hole pair into two transverse modes with a positive full-Hamiltonian probability bound and complete energy, charge and momentum accounting. A declared Gauss Green operator has controlled Coulomb spatial dependence; an autonomous finite current experiment supplies force and displacement records. Two transverse Maxwell modes and negative atomic standing-wave levels are computed with complete source complements. In atomic units, the first excitation gap approaches $3/8$. These constructions reproduce characteristic field and atomic structures on distinct declared sources; their masses, couplings and physical clock maps are not inferred by aggregation [125].

For the complete kinetic/pairing source, the charged native frames obey

$$h_k^2 = \left(\sum_i \sin^2(a p_i)/a^2 + e_k^2 \right) I.$$

This identity supplies the massive wave law on the controlled low-momentum domain while retaining every regulator valley. A portable Taichi stencil executes all 90 canonical fermion modes and

180 doubled coordinates per site inside a pinned OpenWave checkout. Complete 11520- and 38880-coordinate runs agree with independently evolved sparse-source records; the twenty-step numerical record discrepancy is below 6.85×10^{-8}, with an exact-arithmetic integration bound 1.667×10^{-6}. Preparation failures and numerical norm drift are retained. This is executable finite-source reproduction, with no external model award or interacting continuum claim [126].

9 Interacting finite matter and controlled field limits

The reconstructed generator $-\log P$ has operator-norm distance exceeding 0.01984562 from every quadratic in its fixed occupation dictionary; this establishes finite interaction, without a relativistic limit. The executable specification also states instruments, memory and field services; its field actions and coefficients are inputs. Coherent ordering memory realizes the $\mathbb{Z}/2$ exchange-history charge of the native two-load configuration complex; its colored extension has fermionic exchange. Complete field factors then act on the full occupation algebra with a common calibrated clock and retained records. For projector-feedback memory, summing every stationary lag gives a coherent ordering potential and positive diffusion, with a fixed-time $O(h^2)$ unread-channel error bound for pulse step h. This semigroup requires stationary product preparation; it does not identify complete record tapes or physical masses [72, 13].

9.1 A nonquadratic dynamics on the localized algebra

On a fixed localized algebra, the specified parity-even interaction $V = \lambda(N_W - 90)^2$ commutes with all native internal currents. Complete-memory contraction gives unread-state error $O(1/n)$ and complete-history error $O(n^{-1/2})$ for n pulses at fixed time, including arbitrary initial memory correlations. The fixed texture and finite algebra identify no translation-invariant many-particle theory [13].

A faithful Gibbs reference has $S' = \beta(E' - \langle X \rangle)$ and $F'' = -\beta C_\rho(X, X) \leq 0$ for noncommuting X. Projected coupling and hopping forces remain distinct from geometric stress, and Gibbs stationarity does not supply fixed-volume geometric entropy stationarity [10, 13].

9.2 What survives spatial refinement

At fixed spacing the Weyl/contact source has a positive-energy interacting ground and a time-reflection-positive Euclidean hierarchy. With $c = v/a$ and clock $H - UN/2$, the full eight-valley continuum is free for $0 \leq U = o(c^{2/7})$. In the opposite regime $U/c \to \infty$, original charged response leaves every fixed energy window. Holding the cell quartic coefficient a^3U at a positive constant makes that weight $O(a^4)$. These sufficient bounds constrain possible interacting scalings; neither identifies massive Standard Model species [72, 13].

The scaling $c = J|E|^4$, $U = J|E|^8$ gives fixed-spacing interacting Heisenberg dynamics. Infrared bounds supply an ordered ground with $m_s > 1/1000$ and canonical fields with continuous translations, finite propagation and positive gapless energy. No sharp relativistic shell or surviving continuum interaction is proved. Their Gram constructs position conjugate to spatial momentum, with complete finite detectors and ordered source limits. Covariant detectors retain all time-evolved field responses without identifying particles; total-load coordinates are scalar on the protected spin sector. Clocks, spectral measurements and detectors retain their preparation and control premises [72].

At sufficiently small J/B, the declared spin source has an isolated lattice band with equal conjugate internal-charge energies. Its gap and locality bounds force zero physical speed under

local refinement with bounded rest energy and J/B, excluding a massive relativistic limit on that clock and spatial map [13].

At $U = 0$, the ball modular charge of conserved Wick stress is $(2\pi/v)\int(R^2-r^2)\mathcal{E}/(2R)\,d^3x$. Nine independent shell preparations approach the same leading stress response with derivative control. Their relative entropy has an explicit sixth-order coefficient and an $o(R^4)$ nonlinear remainder in the stated lattice/small-ball regime. Responsive area and one common interacting state satisfying geometric stationarity are separate gravity premises [10].

9.3 Two native clocks on one geometry source

The native eight-level energy has an exact representation by three fermionic occupation bits. Two bits serve as independently prepared clocks within the same source cell, on one 80000-state quantum geometry/wave action. Complete joint Ramsey and arrival instruments retain both clocks, their setting coins, source backreaction and no-hit outputs. Their normalized finite-strength vacuum responses differ by 31.37–32.87 parts per million. A certified partial-transpose witness in the vacuum-field controls shows geometry-mediated clock entanglement; this witness is not transferred to the arriving-wave preparation. These finite-source effects do not identify a measured equivalence-principle violation.

For the declared probe family $H_\lambda = H_{\mathrm{env}} + \lambda B(an_a + cn_c)$, both normalized phases converge to the common integral of the Heisenberg-evolved B. For $0 < \lambda \le 10^{-3}$ and source parameter at most 8, their difference is bounded by $1500\lambda/(1 - 600\lambda^2)$, uniformly over environment states and passive references initially independent of the two prepared $|+\rangle$ clocks. This finite Duhamel theorem supplies a common operational weak-probe clock, with no external global time and no physical-second or Einstein proper-time calibration [127].

Positive energy and a shared three-dimensional observation. A separate autonomous source couples the positive native ground-relative energy through $e^{FQ}K$ to a two-state geometry coordinate. Its exact ground-energy mixed derivative is negative for all positive source and probe energies; the complete 49 pairs are tested at three separations. The source is specified in `model/openwave_universal_energy_geometry`.

A complete three-dimensional Maxwell complex has 882 directed edges and 540 transverse coordinates, with Gauss law imposed at every vertex. The declared positive-frequency field, two native clocks and the quantum geometry share one autonomous Hamiltonian. Complete finite records give arrival delay 0.006354 ± 0.000509, transverse displacement -0.003216 ± 0.000050, and Fourier-direction change -0.004595 ± 0.000057, in source units. The joint native clock phase ratios are approximately 0.992837 and 0.992829. Photon vacuum, no-arrival, polarization complements and apparatus work are retained. These are finite constitutive Maxwell records, with no canonical metric quantum-field or Einstein-continuum transfer asserted. A separate attachment of the electron energy and strong length fails both the observed Newton strength and light speed; it supplies no fitted or derived physical G.

The full vacuum-plus-one-photon carrier has 69248 coordinates, including the native source, both clocks and geometry. A zero-bending spatial-response control and the second polarization are retained. The conditional unit map gives $G \simeq 79.293625\,\mathrm{GeV}^{-2}$, about 1.18×10^{40} times the observed strength, and propagation speed $0.0019575c$. These failed unit comparisons apply to the stated energy/length attachment; the strong and geometry sources are not identified by a common-Hamiltonian transfer (`model/openwave_transverse_gravity`).

Common-unit compatibility and collective gravity. For the native-energy geometry with electron unit u and weak coefficient g, the unit equations imply $G_{\mathrm{model}}/(c_{\mathrm{model}}/c) = g/u^2$; changing length alone cannot repair the failed attachment. A distinct member takes $H_g = (\omega/2)\sum_i (I - X_i)$ and $Q_N = (q/N)\sum_i Z_i$. For declared $q = 1/8$ and $\omega = 1$, its coefficient on bare energy labels obeys

$$\frac{g_0}{N}\left(1 - \frac{q^2}{3N^2}\right) \le g_N \le \frac{g_0/N}{1 - q^2/(2N)}, \qquad g_0 = \frac{Sq^2}{2\pi\omega}.$$

Choosing $N \simeq 6.04 \times 10^{42}$ from the observed Newton coefficient makes weak gravity and free-symbol light speed compatible with the unchanged electron anchor. This is hardware calibration to G, with free register gap ω and full coordinate-readout work $N\omega/2$. No common strong/geometry spatial theory or dressed-photon identification is supplied (`model/openwave_gravity_unit_constraint`, `model/openwave_collective_gravity_strength`).

For the complete native geometry source, the actual reference-normalized weak-clock log-lapse derivative is $-2\sinh(q)\tanh(qF)/\omega$, with profile error at most $2\sinh(q)(qF)^3/(3\omega)$. Commuting lattice derivatives give exact linearized Bianchi and gauge identities with every boundary halo retained. Their curvature Laplacian differs from the nearest-neighbour Poisson operator by $(a^2/4)\sum_i \Delta_i^2$. Zero spatial linearized stress forces $\gamma = 1$ in the stated metric family; that stress premise and physical metric identification are distinct from the clock calculation (`model/openwave_source_ward`).

Staggered scalar, vector and tensor parities give exact Bianchi identities and a connected nearest-neighbour scalar graph. Complete three-outcome instruments on the native 60-column source acquire ten noncommuting observable means from independent copies. Their thirty exact probabilities factor the entire 34-site stress/curvature stencil. If every mean has error ϵ and C bounds its coefficient norm, the null-charge error is at most $8C\epsilon$. The sharp inverse $2 + \sqrt{3}$ and explicit variation budget transfer this to tensor and connected scalar-edge errors. Backaction, source work and the 3.23935×10^9-copy sufficient budget for $\epsilon = 10^{-3}$ at joint confidence 0.99 are retained. The metric readout is declared and its nonzero Einstein residual is recorded (`model/openwave_stress_curvature_instruments`).

For this register, Perron positivity reduces the full geometry-sector gap to two spin-sector Jacobi matrices. Exact Sturm inertia, product-state comparison and Hellmann integration control finite-strength mutual forces for all 49 positive native-energy pairs. A complete covariance protocol uses pairs of coordinate records x, y with statistic $D = (x - y)(e^x - e^y)/2$. Its mean is $C = \mathrm{Cov}(Q, e^Q)$ and its relative variance is bounded independently of N. Median block means therefore give an N-independent relative-error copy bound. Pair work is $N\omega u$, while required coordinate resolution scales with $N^{-1/2}$. The same records estimate $M(1) = \langle e^Q \rangle$.

On one common matter/photon extension, weak rest and photon slopes are $M(1)$ and $M(f_\gamma)$, where $M(f) = \cosh(fq/N)^N$. The actual weak-rest normalization gives

$$\frac{G_{\mathrm{weak}}}{c_\gamma/c} = \frac{SC}{2\pi\omega u^2 M(1) M(f_\gamma)}.$$

For a far photon reference, this reduces to $Sq\tanh(q/N)/(2\pi\omega u^2)$. Complete record confidence intervals propagate through this expression. No independently measured native covariance selects N; the count chosen from measured G retains that calibration. The declared extension and physical metric identification remain separate (`model/openwave_geometry_covariance_protocol`).

Full-Haar dynamics on a refined spatial cube. The actual refined cube has 27 vertices, 54 links and 36 faces. Fusion-tree Gauss contractions give a complete 201-mode frame through electric energy 9/2, retaining all 7,236 face columns and every exterior intertwiner. Whole-complement

certificates compare coarse $(a, g^2) = (1, 64)$ and fine $(1/2, 16)$ sources in the same electric clock. The exact Wilson moment $m_1 = \alpha L(1 - \langle W^2 \rangle)/4$, variance v and gap γ give $ve^{-tm_1/v} \le C(t) \le ve^{-t\gamma}$ (`model/openwave_refined_haar_source`, `model/openwave_refined_haar_spectrum`).

Energy-dependent full-complement Schur bounds give coarse and refined gaps [96.00000060968, 96.00000069820] and [47.99403921427, 48.00529277681] in the same source clock. For three subdivisions $L = 1, 2, 3$, a fixed physical unit-square loop provides a local refinement test without an extensive low-energy projector. With $\alpha = 32/L$, $\beta = L^3/32$, its fundamental and spin-one free frequencies are exactly96 and 256. The full-Haar Duhamel bound is

$$\|\tau_H^t(A) - \tau_E^t(A)\| \le \delta_L = \min\{2, 2\beta N_{\mathrm{inc}}|t|\}, \quad N_{\mathrm{inc}} = 5 \ (L = 1),\ 8L - 4 \ (L \ge 2).$$

Both normalized real correlations exceed 0.97 at common native time 1/4096. Complete three-outcome temporal instruments include backaction; two readouts cost at most 32/3 or 512/27 source units. Every Haar representation remains. The error grows as L^4 at fixed time, so the estimate supplies no continuum or Clay gap (`model/openwave_local_gauge_memory`).

Positive heat-kernel transfer gives time-reflection positivity and the strong Hamiltonian-clock product limit. At sufficiently strong coupling, the electric-star bound $64/(g^2)^2$ admits Yarotsky's stability theorem, giving a uniform-volume gap at fixed spacing with existential constants. Whole-lattice local propagation has explicit norm-convergence bounds; neither result is a four-dimensional continuum theorem (`model/openwave_su2_transfer`, `model/openwave_su2_volume_gap`, `model/openwave_su2_propagation`).

Coupling response and actual spatial blocking. For the full-Haar shared-link source and $x = g^2$, the identity $H'(x) = H(x)/x - 4(2 - W_1 - W_2)/x^2$ reduces gap derivatives to bounded Wilson expectations. Holding $\Delta(x(a))/a$ fixed requires $dx/d\log a = \Delta/\Delta'$; this is a necessary source-clock tangent, not an inferred physical beta function (`model/openwave_haar_coupling_flow`).

Subdividing the spatial cube gives 54 links and 36 faces. Ordered fine-link products embed the coarse Haar/Gauss space isometrically. Exact conditional integration gives

$$\mathbb{E}[e^{\kappa \sum_{36} W_f} \mid U_c] = 1 + \tfrac{9}{2}\kappa^2 + \tfrac{321}{32}\kappa^4 + \tfrac{\kappa^4}{64}\sum_6 W_c + R_\kappa,$$

$$\|R_\kappa\|_\infty \le e^{36|\kappa|}(36|\kappa|)^6/720.$$

Complete face parity and an independent Haar contraction certify the positive coarse Wilson coefficient. Electric compression doubles the coarse Casimir sum; matching it at half spacing in the same clock requires $g_f^2 = g_c^2/4$. The magnetic mismatch and induced-operator remainder preclude identifying this step with full-Hamiltonian renormalization (`model/openwave_gauge_refinement_clock`). At fixed graph and coupling, the positive-time Euclidean response collapses as $a \to 0$. No physical GeV glueball or nontrivial four-dimensional continuum is supplied. Exact flat-vacuum dilation covariance also excludes a positive gap in a theory with a nonzero finite-energy excitation; Yang–Mills does not impose that covariance (`model/openwave_euclidean_refinement`).

10 The Einstein equation from a realized null balance

The gravitational hypothesis identifies Einstein dynamics with the coarse response of the same process.

Theorem 10.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 7.2; nothing below*

depends on the choice. If a symmetric form X satisfies $X(k,k) = 0$ for every null vector k, then $X = \phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1, \pm e_i)$ and $(1, s, s, \pm s)$, $(1, s, -s, s)$ with $s = \sqrt{3}/3$, reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ with design determinant $8192/27$ in the certificate's basis and a decoder obeying $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$.

Proof. On future null rays $(1, n)$ the identity $X_{00} + 2X_{0i}n_i + X_{ij}n_in_j = 0$ on the unit sphere forces $X_{0i} = 0$, $X_{ij} = \phi\delta_{ij}$ and $X_{00} = -\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 10]. □

The same decoder gives the sharp entrywise error bound $(2+\sqrt{3})\epsilon$ for null-charge errors at most ϵ. A Ward residual w bounds the remaining metric coefficient by $|c(x+e_j) - c(x)| \le aw + 8(2+\sqrt{3})\epsilon$ for forward differences. Centered derivatives control only two-step differences; the exact checkerboard counterexample forbids substituting nearest-neighbour connectivity. These finite error inequalities are machine-checked and executed on native-profile curvature, retaining the undetermined vacuum metric term (`model/openwave_einstein_error`).

Theorem 10.2 (Tensor completion of a realized null balance)**.** *Let (M, g) be a connected, time-oriented, C^3 Lorentzian four-manifold and T_{ab} a symmetric C^1 tensor with $\nabla^a T_{ab} = 0$. Suppose a common realized tower supplies $\kappa = 8\pi G_N > 0$ with*

$$G^{(g)}_{ab}k^ak^b = \kappa T_{ab}k^ak^b \tag{13}$$

for every null vector k at every point. Then there is a constant Λ with

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}, \tag{14}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ fixes $\Lambda = \Lambda_0$.

Proof. Theorem 10.1 applied pointwise to $G^{(g)} - \kappa T$ gives $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b\phi = 0$, connectedness makes ϕ constant, and $\Lambda = -\phi$. □

Null balance fixes Einstein's equation up to a metric term; the Bianchi identity and stress conservation make its coefficient the cosmological constant. The coefficients of entanglement-equilibrium arguments [65] enter through one machine-checked identity: scalars with $\delta S_{\mathrm{bulk}} = \frac{8\pi^2\ell^4}{15}t$, $\delta A = -\frac{4\pi\ell^4}{15}f$ and $\delta S_{\mathrm{bulk}} + \delta A/(4G_N) = 0$ obey $f = 8\pi G_N t$. For OPH to supply (13), one common refinement tower must realize (G1) the effective manifold of Proposition 7.6; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L\rangle = \delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa = 8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

Section 9 supplies sector-relative modular and Ward inputs to (G2)–(G3) for the proved free Weyl sector, and the matter first-law and small-ball parts of (G4)–(G5). Nine explicit shell perturbations span traceless stress variations; finite boosted-ball responses have full rank for $0 < R/L \le 2^{-28}$. At exactly $U = 0$, compact lattice states and quadratic detectors recover the leading stress response in a joint spacing/small-ball limit. On each fixed ball, the actual finite-density modular flows also

converge on all fixed probes, with a generator graph core and convergent selected relative entropies. The proofs retain all eight valleys and do not assume a uniform covariance gap [73]. The Wilson source supplies nonzero energy-current response to a finite pulse, with complete records and positive work. Its energy Ward identity retains applied work. The regional modular first law holds on supported marginals. A physical comparison bounds correlations with the polynomial-state remainder. Geometric stress, uniform area response and universal coupling with calibrated G_N require source maps [10].

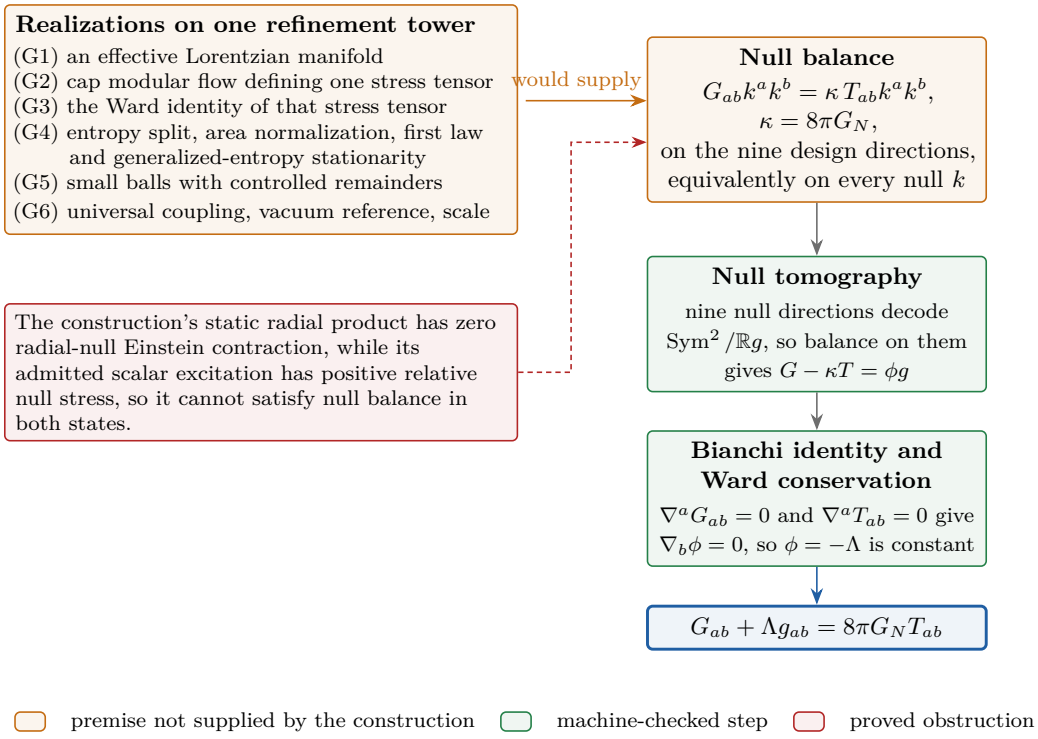


Figure 3: The Einstein equation as tensor completion of a realized null balance (Theorem 10.2). The tensor step is machine-checked from an explicit null-balance premise. Realizations (G1) to (G6) on one refinement tower would supply that premise, and no complete realization of them on one common refinement tower is established; the tested static radial product fails the balance for its own scalar excitation.

11 Cosmology and the closure coordinates

The cosmological hypothesis asks the same process to fix the vacuum, the dark sector and the primordial response. In this paper the layer consists of conditional branches and diagnostics of declared maps.

11.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular

charge, with anomalous modular energy on overlap collars as one candidate carrier [16]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit counterexample. A per-cut model gives $a_0 = Gn^2 c$, with only the product $n^2 c$ identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\mathrm{DE}} = \kappa/N$ at constant κ, with the separately conserved Friedmann continuity law, gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [17].

11.2 Expansion, horizons and records

Expansion adds no causal data to the layered read law. A physical read $\|\sigma_{j+1}t - \sigma_{j+1}s\| \leq \sigma_{j+1}a$ holds exactly when $\|t - s\| \leq a$, so every scale-factor profile generates one order, as conformal invariance of causal structure requires [46, 47]. The profile enters only the count measure: a constant physical event density gives the four-volume weight $\sigma^4\, d\eta\, d^3x$, and the source law does not select σ [18]. On a finite screen of fixed capacity, maximizing generalized entropy over sector probabilities gives $S_{\mathrm{gen}}^{\max} = \log M$, and uniform transfer of a fraction f of the horizon capacity to an observer changes that maximum by $\log(1 - f) < 0$, the sign of the de Sitter time advance [15]. Public records keep an exact information ledger: pinching is an average of independent sign unitaries whose entropy gain equals the relative entropy to the pinched state, and on a supplied three-dimensional layered population crossing reads obey an area law with coefficient $(\pi/4)n^2a^4$. A declared nat per crossing calibrates that area; horizon entropy, Hawking flux and evaporation need the native channel and a physical geometry [19].

11.3 Native populations and thermal lattice comparisons

The native runs cover a finite number of repair ticks. Reading a tick as $\hbar/E_{\mathrm{cell}} = 6.885 \times 10^{-44}$ s requires a supplied energy and clock dictionary. Every defect class appears in pairs on the first consistent slice, left-handed quark classes annihilate, the other classes freeze once the loads balance, and class and conjugate stay balanced (Section 8.6). Of all defect identities, the two left-handed quark classes take 24.1 percent each, the two right-handed quark classes 19.6 percent each and the lepton class 12.7 percent, the same within 0.3 points from level six to level ten. Equal occupation of the Standard Model states of each class, as in a hot plasma, gives 21.4 percent for each quark class and 14.3 percent for the lepton class; the runs lie within three points of these values and differ by about a tenth, in the order of the register's declared costs. These percentages compare register classes with an assumed equal-state Standard Model census; they are not measured early-Universe particle abundances. Separate Euclidean lattice ensembles define $T = E_{\mathrm{cell}}/N_t$ using their supplied scale dictionary.

Three conditional comparisons use declared continuous-link actions. With the couplings of the declared chain and the published reduction to three dimensions [20], the electroweak central crossover temperature is 161.1298 GeV; the Standard Model lattice comparison is 159.5 ± 1.5 GeV [21]. The available marginal variances bound the temperature standard deviation by $[0.11810, 0.13680]$ GeV under independence, or $[0, 0.27631]$ GeV without it. Paired peak replicas are absent, and a separate one-percent allowance has no supplied probability law. With the assumed Standard Model radiation census $g_* = 106.75$, a Friedmann conversion gives about 9.0×10^{-12} s. Di-

viding by the supplied phase-unit duration gives about 1.3×10^{32} units; this is not an elapsed native repair count. Restricted to colour, the seam cost $1-\mathrm{Re}\,\mathrm{tr}_5\,U/5$ is the Wilson plaquette action, the colour centre is the flux classes 2 and 4, and the quark classes are the charged ones. At $N_t=4$ these links deconfine in a first-order transition at $\beta_c=5.69105(39)$ with $T_c/\sqrt{\sigma}=0.614(18)$; the published values for the same theory are 5.6925(2) and 0.598(3) at this lattice spacing [22]. Quenched Wilson quarks on the same links bind: the static potential rises linearly, $m_N/m_\rho=1.45\pm0.13$ against the measured 1.21, and the binding melts above the transition.

The colour and hadron numbers are those of SU(3) gauge theory with the Wilson action, which the model's colour links are once continuous links are declared; their agreement with published lattice values confirms that identification. The crossover takes the chain's declared couplings as inputs. The baryon excess and the light-element abundances are not computed [13, 1].

Capacity work and horizon-dependent expansion. For the finite active-register coupling $H(V,k)=\epsilon V/k$, the operator pressure is $P=-\rho=-\epsilon/k$ at fixed capacity. Complete register/geometry observations retain preparation work and every capacity-change stroke; the four-stroke work cycle vanishes exactly. Capacity drift has exchange $Q=-\rho\dot{k}/k$, which distinguishes mechanical pressure from the separately conserved effective-fluid pressure. These identities and all finite records are specified in `model/openwave_capacity_stress`.

The proved capacity drift law gives $w=-1+\frac{1}{3}d\log N/d\log a$. With an area capacity $N\propto R^2$, a comoving radius gives $w=-1/3$. A Hubble radius with separately conserved dust instead gives $w=0$ and $H^2/H_0^2=a^{-3}$. Retaining mechanical pressure $-\rho$ on that branch requires matter exchange and a constant dark fraction, with no transition from matter domination to acceleration.

For the declared future-event-horizon member with coefficient one, $x=\sqrt{\Omega_{\mathrm{dark}}}$ obeys the exact algebraic equation

$$\frac{x^6(1+x)^3}{(1-x)(1+2x)^8}=a^3\frac{x_0^6(1+x_0)^3}{(1-x_0)(1+2x_0)^8}.$$

Monotonicity and rational bisection replace an ordinary differential equation solver. The future limit has positive finite H, and the event-horizon boundary condition follows. With the supplied $H_0=67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}$ and $\Omega_m=0.315$, all fifteen correlated primary chronometer rows give $\chi^2\simeq6.57717$. The same comparison gives 6.19134 for fixed capacity, 21.54903 for comoving area and 70.48705 for separately conserved Hubble area. The alternative stellar-population reductions use the same observations. These are conditional capacity-history comparisons; neither the cosmic amplitude nor a physical register-to-horizon map is derived. The construction and complete records are specified in `model/openwave_horizon_capacity`.

With conserved ideal radiation, this event-horizon branch obeys $d\Omega_x/d\log a=\Omega_x(1-\Omega_x)(1+r+2\sqrt{\Omega_x})$, where $r=\rho_r/(\rho_m+\rho_r)$. Outward Taylor integration reaches $z=10^6$. Its early radiation fraction scales as a^2, compared with a^4 for fixed capacity and a constant fraction for the Hubble-area control. Present amplitudes and radiation content are supplied; the calculation derives neither particle abundances nor an independent cosmic amplitude (`model/openwave_radiation_capacity`).

The adiabatic barotropic attachment has $c_a^2<-1/5$ throughout the supplied past domain. The established holographic-quintessence reconstruction instead has $\phi'=M_{\mathrm{Pl}}\sqrt{2\Omega_x(1-x)}$, $V=\rho_x(2+x)/3$ and $c_s^2=1$. Exact radiation-inclusive Bernstein bounds, checked in Lean, give $2\le V''/H^2\le6$. This completes a local scalar background model, without deriving a particle mass, a native-source scalar or a perturbation likelihood (`model/openwave_capacity_perturbations`).

For a declared Einstein–scalar–dust–radiation attachment, the exact constraint obeys $C_N=-(3+2h)C$. The regular zero-wave-number mode reduces to one positive quadrature and gives

present metric coefficient $[0.73417927, 0.73425730]$, against $[0.70910831, 0.70918107]$ for the same-density constant-Λ control. A separate preparation at $N = \log a = -1$ has an explicit dust correction enforcing the full finite-k constraint. Comoving variables remove the exact zero mode; an eight-coordinate interval transport and complete k^4 remainder certify its finite-wavelength evolution to $N = 0$. The metric k^2 coefficient lies in $[-0.008281806997, -0.005759370982]$ and the comoving-dust coefficient in $[-1.455879943233, -1.454618725225]$. This is a late-time supplied preparation, without a primordial continuation or CMB comparison. The scale symmetry $H_0 \mapsto \lambda H_0$, $\rho \mapsto \lambda^2 \rho$, $L \mapsto L/\lambda$, $k \mapsto \lambda k$ retains the absolute normalization premise (`model/openwave_capacity_finite_wavelength`).

The covariance and shared chronometer data are from Moresco et al.; the supplied cosmological inputs are from Planck2018VI.

11.4 Pixel closure and the common particle response

The dimensionless pixel coordinate is $P = a_{\mathrm{cell}}/\ell_P^2$. Its self-reference equation relates it to a response computed from the chosen source specification,

$$P = \varphi + \frac{\sqrt{\pi}}{A(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2},$$

where A is the declared inverse electromagnetic-coupling readout. A fixed-point proof determines a root of that map. Identifying its readout with the physical Thomson coupling requires the current response of the same action, vacuum, spectrum and clock.

One explicit conditional map joins the native matter census to the charged-lepton law of Section 8. For unified coupling u, energy unit E_* and scale coefficient β, take

$$\frac{M_U}{E_*} = e^{-2\pi} P^{1/6}, \qquad \frac{v}{E_*} = P^{-1/2} e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi} \log \frac{M_U}{m_Z}.$$

With $\beta = 4$ the exponent is the plaquette cost of one weak-centre flux quantum. The central element $\mathrm{diag}(1,1,1,-1,-1)$, which is also the 2π rotation of Section 8.6, has the shortest generator $H = \mathrm{diag}(0,0,0,i\pi,-i\pi)$ with $\mathrm{tr}_5\, H^\dagger H = 2\pi^2$, so $2\pi/(4u) = \mathrm{tr}_5\, H^\dagger H/g^2$ at $u = g^2/4\pi$; the colour and generating central elements give $\beta = 3$ and $12/5$ [11, 1]. The infinite heat-sum equation, with certified tail bounds, supplies u; the same neutral-Higgs relation supplies m_Z/v. Exact charge and Casimir traces on the fifteen native matter rows, with three families and one complex Higgs doublet, give

$$(b_1, b_2, b_3) = (41/10, -19/6, -7)$$

when inserted into the declared four-dimensional one-loop functional. The representation traces are finite consequences of the gauge module; the loop functional and scale law are additional physical modeling inputs. This is the intended surviving chiral census; applying it to the eight-valley lattice requires a proved mode-removal or threshold map [12, 11, 13].

For every charged lepton and all six quarks, the same coefficient masses enter the response kernel

$$K_f = \frac{2N_c Q_f^2}{\pi} \int_0^1 x(1-x) \log\left(1 + \frac{m_Z^2}{m_f^2} x(1-x)\right) dx,$$

$$A = \alpha_2^{-1} + \frac{5}{3}\alpha_1^{-1} + \sum_\ell K_\ell + \left(1 - \frac{3\alpha_3}{\pi}\right) \sum_q K_q.$$

The quark multiplier is a prescribed screening factor. This map includes neither complete charged-vector and scalar response nor a nonperturbative hadronic spectral measure, so it is not the full physical electromagnetic running of the Standard Model.

The minimal census with $\beta = 4$, positive couplings, $P \geq 1$, $M_U \geq m_Z$ and no intermediate thresholds has no solution: the scale equations force $\alpha_2, \alpha_3 > 1/6$, whose combined heat sum is below $1/80$, while pixel closure requires at least $1/4$. With the explicit alternative $\beta = 7$, interval bounds certify a locally unique self-consistent root

$$P \simeq 1.628533369611833, \qquad A \simeq 168.815082928453.$$

Infinite heat tails, implicit-root derivatives and a strict interval self-map are controlled. The memory inverse is certified over the same root enclosure, so its two admissible controllers realize those same charged-lepton coefficients. This is a consistent conditional coefficient map; its inverse coupling is not an identification with the measured fine-structure constant [13].

The specialist coupling analysis also evaluates supplied running slopes, thresholds and scale forms. Agreement obtained by selecting those inputs does not derive the current response of the native interacting source. The source census and its regulator multiplicities determine which running functional is applicable; the physical reduction cannot be replaced by a more favorable census [11, 13].

A complete physical loop must replace the coefficient readout by the current spectral response of the actual interacting states. Compact binding energies and quartic interactions change that spectrum. Therefore closing the displayed scalar map does not close the joined physical particle-response loop. Detector use of $g(P) = \sqrt{4\pi\alpha(P)}$ reads a supplied coupling; it does not independently derive its electromagnetic meaning.

A separate exact scalar is due to Wissner-Gross [24]. For four slots of dimensions $(3, 2, 3, 2)$, traceless letters on unordered slot pairs give 169 product records and three exchange-odd directions complete a 172-dimensional feature space. Two commuting signed exchanges X, Y define $K = (2I - X)(3I - Y)/2$, whose joint projectors have gains $1, 2, 3, 6$ and ranks $79, 48, 42, 3$. With the declared normalization 31104,

$$\alpha_{\text{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\text{pair}}^{-1} = 137.035999189566\ldots,$$

against the measured 137.035999177(21) [25]. An independent reconstruction verifies the arithmetic with no continuous coefficient. Its identification with the Thomson coupling, its normalized-trace readout and its attachment to the screen grain are conditional [11].

The other closure coordinate, record capacity N, bounds the declared correctable record algebra. Under its de Sitter entropy interpretation,

$$N = \frac{3\pi}{\Lambda \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}. \tag{15}$$

This dictionary converts a capacity into a cosmological constant after physical scale calibration. It does not make a fitted capacity an independent cosmological prediction. The specialist analyses separate conditional root certificates, measured-coordinate comparisons and independent physical tests [11, 17].

12 Realized results and physical obligations

The physical target is one realization whose limiting observable laws agree with measurement. Finite certificates establish their displayed domains; calibration and physical identification are

separate statements. Table 3 collects the common-source requirements and the negative controls that delimit the constructions. Existence and phenomenal experience are interpretive questions: an implemented record interface does not derive why anything exists or the phenomenal character of a record. The construction and field papers give the detailed premise and artifact maps [72, 89].

Table 3: Constructed domains and their physical extension. The stated failures apply to their tested member or identification.

Domain	**Constructed result**	**Required extension and negative controls**
Observers and records	Finite consensus normal forms, record-caused feedback, complete instruments, quantum identities and repair entropy laws	The declared preparations, algebra, scheduler and resource accounting must belong to the same process. Classical committed records alone supply no Bell violation; consensus does not select a unique physical history.
Space and fields	Exact rank-three carrier response, causal event records, controlled free-field limits and specified interacting finite sources	Physical position, common clocks and an interacting chiral relativistic limit require realization. Full-port refinement loses the rank-three band; the screen spectral growth does not identify an elliptic three-manifold.
Gauge and matter	Conditional Standard Model Lie type, a declared matter representation, centre characters and native current operators	Centre classes do not identify species. Charges, spin, statistics, mixing, and interacting particle poles must be recovered on one source and one compatible physical limit.
Masses and decays	Conditional lepton ratios, calibrated hadron energies, propagating finite bands, complete weak records and calibrated neutral oscillations	The muon discrepancy, failed neutral-pion and charge-radius comparisons, and unused strange and helium masses are retained. Calibration targets are not independent predictions; an open energy shell supplies no physical decay amplitude by itself.
Gravity	Null tomography and tensor completion, matter-energy identities, modular work bounds and shared photon/clock geometry records	The metric, conserved stress, generalized entropy and clocks must realize one refinement tower. The tested static scalar member fails its null balance; the specified physical G and light-speed attachment fails.
Cosmology	Capacity-work identities, conditional expansion histories, the deep-law implication and public-data comparisons	Vacuum, horizon and physical-scale selection are explicit. Shared-data alternatives are not independent confirmations; the tested minimal-matter pixel roots do not reproduce the measured coupling.

12.1 The realization maps

Five maps specify physical identification independently of whether the finite mathematical result uses them.

(M1) Position and scale. A physical position and length for the carrier completion, compatible overlap gluing and a cofinal spatial refinement (Section 5).

(M2) Clock and energy calibration. One comparison between internal evolution, laboratory duration and measured energy. The calibrated reference clock and discrete native event counts are distinct; the rejected direct count-to-phase attachment does not reject the reference-clock construction.

(M3) Laboratory current. A physical gauge current and matter action with a continuum operator limit for the finite port response.

(M4) One common physical tower. One refinement family realizing the event, modular, stress, entropy, continuum, vacuum and scale premises together.

(M5) An implemented and matched action. A field and interaction law implemented by the same native services with complete states, relative normalizations and readouts. Declared architecture and empirical calibration are distinguished; physical propagation, currents and interactions must have compatible limits.

The finite sources realize restricted instances of these maps. Their composition into the required physical theory does not follow from separate successful instances. A failure of a tested attachment rejects that attachment; a framework-wide rejection additionally requires a proof that it is forced.

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [4]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer. Published records also test precise attachments of native records retrospectively.

13.1 Matches with nature

The project keeps a public comparison inventory [2]. Its entries include structural identities, finite-source phenomena, numerical comparisons, calibration targets and open items; they are not a count of independent observational validations or of upstream OpenWave awards. OpenWave lists OPH as model M10 with all 31 criteria untested. Its admission rule requires the native carrier process to generate the tested field records; installed source actions do not earn those criterion validations. Structural results include the conditional Standard Model gauge Lie type and a declared matter representation with eight colour generators, one Higgs doublet, the generation charges and hypercharges, three colours, left-handed weak couplings and conserved $B - L$. Finite-source results include a three-dimensional slow response, causal bounds, a Dirac walk limit, spin $1/2$, Coulomb response, two transverse light polarizations, discrete atomic levels and three charged-family branches. Their source domains and physical-identification conditions remain separate. Table 4 retains the numerical comparisons, including their nonzero discrepancies.

The Landau and Aharonov–Bohm rows compare the declared CF-1 routing walk with continuum formulas. The electron g is a finite dressed-current response, not a precision derivation of the anomalous moment. The electroweak row uses the declared coupling chain and imported reduction, not the actual one-doublet source inputs. The colour transition is a same-action pure $SU(3)$ Wilson benchmark at one lattice spacing; the nucleon-to-rho ratio comes from quenched Wilson

Table 4: Numerical comparisons on the project's results page [2]. References include measurements, continuum formulas and lattice benchmarks. Relative deviations alone do not establish agreement within experimental uncertainty. [a]The measured electron mass sets the unit; the lepton ratios, masses and Koide statistic share one retrospective construction and are not independent tests. [b]Declared coupling chain, distinct from the native one-doublet inputs. [c]Against the joint ACT DR6, Planck and DESI chains the value lies in the lower 0.6 per cent tail (Section 13).

Quantity	Source	Reference	Diff.
m_μ/m_e	206.770316	206.7682827(46)	+0.00098%
m_τ/m_e	3477.4728	3477.365(176)	+0.0031%
m_τ/m_μ	16.818047	16.817692(852)	+0.0021%
Koide Q	2/3	0.6666645(51)	+0.00033%
m_μ (MeV)[a]	105.659414	105.6583755(23)	+0.00098%
m_τ (MeV)[a]	1776.985	1776.93(9)	+0.0031%
v (GeV)[b]	246.60	246.22	+0.15%
$\alpha_s(m_Z)$[b]	0.11834	0.1180(9)	+0.29%
$\sin^2\theta_W^b$	0.23074	0.23129(4)	−0.24%
$1/\alpha(m_Z)$[b]	128.308	127.930(8)	+0.30%
$1/\alpha$[b]	136.995	137.035999177(21)	−0.030%
m_Z tree (GeV)[b]	91.588	91.1880(20)	+0.44%
m_W tree (GeV)[b]	80.330	80.3692(133)	−0.049%
m_H (GeV)[b]	125.04–126.02	125.13(11)	−0.07–0.71%

Quantity	Source	Reference	Diff.
τ leptonic ratio	0.97256384	0.9730(22)	0.20σ
Electron g	1.99828	2.00232	−0.20%
Landau level/$\sqrt{2B}$	0.991	1	−0.90%
Aharonov–Bohm	0.980	1	−2.0%
EW crossover (GeV)	161.1	159.5(1.5)	+1.0%
$T_c/\sqrt{\sigma}$	0.614(18)	0.5982(34)	+2.6%
m_N/m_ρ	1.45(13)	1.211	+20%
n_s^c	0.9660	0.9649(42)	+0.12%
$dn_s/d\ln k$	0	−0.0045(67)	0.7σ
r	0	< 0.036	bound
Ω_K	0	0.0007(19)	0.4σ
f_{NL}	0	−0.9(5.1)	0.2σ
Isocurvature	0	$< 2.5\%$	bound

quarks. These are distinct source comparisons, not a common-source physical thermal history. The primordial entries are conditional branch outputs and shared-data comparisons, without a joint independent likelihood.

Nine further entries are calibrated strong-source fit targets: the proton, neutron, charged-pion, ρ^0- and Δ^+-assigned energies, the axial magnitude, the kaon, the Lambda and the triton. Reproducing these supplied targets is not an additional independent mass prediction. Partial results include gravitational attraction without a derived G and the cosmological constant, $1.096 \times 10^{-52}\,\mathrm{m}^{-2}$ against 1.089×10^{-52} with the capacity chosen in view of the measured value. Open items include the source's own hypercharge slope ($\sin^2\theta_W = 0.2056$), independently identified quark and neutrino spectra and mixing, the absolute lepton scale, the matter excess over antimatter and the Tully–Fisher exponent of galaxies, 4 against 3.75 ± 0.09.

13.2 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [60].

Theorem 13.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1 - \cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order a^4k^6, suppressed by exactly $1/16$ under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 13.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 13.1 with scale-independent coefficients, the complete orbit of the twelve port directions u_i is its sole hop support through order k^6, the carrier scale a is finite and positive with the quadratic term normalized to k^2, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}[1-\cos(ak\,u_i\cdot\widehat{k})] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{16}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$, so in a transported frame the branch predicts the scale-free relations

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{17}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ vanishes.

Proof. The exact carrier moments $\sum_i(u_i\cdot n)^2 = 4$, $\sum_i(u_i\cdot n)^4 = 12/5$ and $\sum_i(u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 13.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module D_6 of Theorem 5.1, with Smith invariants $(1,1,1,1,1,2)$, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight $1/60$ on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Big[1-\cos\Big(ak\,w_j\cdot\widehat{k}\Big)\Big], \tag{18}$$

with $C_4 = -a^2/20$, $B_0 = a^4/840$ and $B_6 = -a^4/12600$, hence $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$ and $B_6/B_0 = -1/15$. The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated $\mathcal{I}_6$ orbit.

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 5.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j \cdot n)^2 = 10$, $\sum_j (w_j \cdot n)^4 = 6$ and $\sum_j (w_j \cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (18) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 13.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits O_s of the sixty proper carrier rotations with radii $r_s > 0$, weights $w_s > 0$ and the k^2 coefficient normalized to one, write $\Lambda_{\mathrm{norm}} = k^2 + C_4 k^4 + (B_0 + B_6 \mathcal{I}_6(\widehat{k}))k^6 + O(k^8)$ and $\mu_m = \sum_s w_s |O_s| r_s^m$. Then $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ for every member; $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \geq 10/21$ with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and B_6/B_0 lies in $[-16/135, 16/75]$, the face and vertex orbits attaining the endpoints and the edge orbit sitting at $-1/15$.*

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_i W_j r_i^2 r_j^2 (r_i^2 - r_j^2)^2 \geq 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21}C_4^2$, $B_6 = \frac{32}{315}C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [4].

13.3 An instrument test from the executable observer

Under a common current-record interface every admissible record law has the classical statistics of C and E. The multiplier g, the coherence between an empty seam and a waiting single load, separates C, E and the law that records only the current. On a fixed coherent two-state source preparation the readout probability is $3/8 + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$: reversing the phase moves C from $5/8$ to $1/8$, E predicts $3/8$ for both, and the law that records only the current predicts $3/8 \pm \sqrt{2}/4$ [74]. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of $1/64$ and an exact sampling certificate below $1/1000$; it rejects malformed or incomplete records and reports

no data when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [72]. A sequential design on one continuing carrier needs neither exact resets nor independent trials: 379 discarded native attempts before each of 3,927 trials replace the reset, failed heralds count as trials, and conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model [93]. Synthetic controls exercise the decision rules and establish no physical verdict; physical identification requires an independently calibrated encoding and comparison data.

13.4 Native records against public data

Published records can test precise attachments of native records without a new laboratory realization; each test below is retrospective, states its readout hypotheses, and assigns no prospective verdict.

The diagonal native branch is a classical stochastic process, so under independently chosen settings and local factorized responses it obeys the CHSH bound $S \leq 2$. The deposited records of the loophole-free Hensen experiment [96] reproduce 245 eligible trials, 196 wins and $S = 2.42250$, and the published fixed-count analysis gives a tail bound near 0.03917 with its setting-calibration allowance [94]. This rejects the local classical reading of the diagonal branch under that experimental contract. Committed native records form a classical process: the native grammar keeps diagonal inputs diagonal under every classical feedback history, and two implementations with the same committed branches, which give 3/4 and 1/4 under a read before commitment, give 1/2 for both once the record is committed [102]. Correlations beyond $S = 2$ therefore need uncommitted coherence from declared preparation services, and every admissible record law attenuates it [74]. The replay leaves the completed record surface untouched, on which a declared entangled state reaches the Tsirelson bound of Theorem 4.5, and it does not distinguish C from E.

Event clocks need no external validation, and identifying their readings with a published measurement is a testable hypothesis. Attaching the published 300-second blocks of the Ar^{13+}/Yb^{+} optical frequency ratios of King et al. [97] to the normalized incident-repair count ratio of two native ports or regions, with a declared gain and frequency bracket, gives a complete-count probability below 10^{-104} for two ports and 10^{-22} for every pair of the 3,433 regional event sets [95]. The direct attachment is rejected together with its readout assumptions; the native event clocks are unaffected as clocks of the model. Published trapped-ion records separately test a supplied oscillator model executed through the source interface [73].

13.5 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 11.4 are diagnostics of declared maps. The registered conditional primordial branch $n_s = 1 - P/48 = 0.96602$, at the calibrated grain P, lies at the lower 0.612 percent position of the official ACT DR6, Planck and DESI joint posterior chains, whose mean is 0.9743 ± 0.0034 [98, 95]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure

of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and algebraic results retain their stated domains under all three.

14 Machine verification and reproducibility

A finite theorem is a deduction from stated premises on a specified domain. An exhaustive certificate belongs to that proof only when its enumeration and soundness argument are supplied. Exact executable certificates recompute integer or rational identities and outward error bounds; numerical receipts retain their declared tolerances. A conditional analytic theorem consumes a specified continuum object, whereas a source limit theorem derives a limit from a declared execution family with explicit normalization and error control. Neither establishes its own physical identification. Reconstruction implications retain their geometric premises and unsupplied realization maps.

Calibration consumes comparison data. A prospective conditional test fixes its observable, precision, decision rule and source before an eligible comparison; a frozen branch prediction additionally identifies which physical branch the result can reject. These distinctions apply to each numerical comparison and prevent the same supplied target from being counted as independent evidence.

The Lean dependency closures reported here contain no admitted propositions. The executable certificates bind their source, inputs and full outcomes, including failed preparation and complementary sectors. Independent replay checks canonical exact outputs and the stated numerical tolerances, while adversarial tests exercise the acceptance boundaries. For the nuclear source, exact Casimir/parity projectors and a positive full Gram matrix certify complete state coordinates; fixed trial vectors and residual bounds then certify the same complete spectra without a floating rank decision.

The source, receipts and rebuild commands are public [1, 3, 73]. The construction paper gives the observer and resource contracts [72]; the source-limit paper states convergence constants and orders of limits [89]. Public-data reducers bind their datasets, observation hypotheses and decision rules [94, 95]. Exact finite certificates, analytic limit proofs and physical comparisons retain these distinct domains throughout the paper.

15 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [28, 29, 30]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion and a positive functional reconstructed from native repair histories. Quantum Darwinism locates classical objectivity in redundant environmental records [55], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [56], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [57], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [53] and the thermal-time programme reads time from modular structure [54]; OPH attaches public facts to protected overlap records and treats modular flow as state- and event-relative. The interacting 384-state native output admits a declared

two-configuration M_2 corner with conditioning probability 0.01191417158 and modular frequency 3.1396822092. Its corner was chosen after inspection of that known output, not as a uniquely selected physical observer. Four ordered projective readouts and the 0.9880858284 outside outcome retain their full poststates and native continuation. The modular parameter is an injected apparatus setting; the corner is not invariant under H, and no native-clock or subjective-duration conversion is established (`code/openwave_candidate/next/C11/INTERACTING_MODULAR_OBSERVER.md`). The local-diamond proof sits beside asynchronous agreement [49, 50], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [51].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [64, 65, 68], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [61, 62, 63]. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [66, 67]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [70, 71] and contributes the forcing of the local Lie type from complete response and rechartings inner on that response, a stronger premise than mere internal availability of their implementers.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [104]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [105], and Nozick proposed that a fundamental principle might subsume itself [106]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [107]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

16 Conclusion

OPH proposes one closed inhabited history whose observers recover the rules of their world and realize those rules inside it. Protected overlap records supply its notion of a public fact. The proposal brings questions of existence, observer access and physical law into one specification, with a clear acceptance condition: the same source must reproduce the observed joint laws under intervention. The finite closure theorem gives an exact model of that logic on consensus packets; a universe-level fixed point is a further claim. The operational observer specifies access and records, without a theorem about phenomenal character.

The finite mathematical results connect repair, quantum records and field observables. Repair gives canonical public normal forms and complete-tape entropy laws. On a supplied complex algebra-state surface, completed records obey the quantum measurement identities. The three-dimensional slow response and conditional Standard Model gauge Lie type coexist with an exact centre-character dictionary. That dictionary does not distinguish every particle species.

The charged register gives $Q = 2/3$ and ratios 206.7703 and 3477.47, with its retrospective

selection ancestry and muon discrepancy retained. A native action fits nucleon/pion energies and the axial magnitude without rescaling the current; the seven-input hyperfine member also fits nuclear binding and rho/Delta-assigned energies, retaining its failed neutral-pion and radius comparisons. Complete spatial complement bounds certify calibrated interacting nucleon source poles, their preparation weights and a positive common kinetic rule. A local strange source has an energy-conserving proton/pion shell, without an executed weak lifetime. Strange and triton calibrations retain failed unused strange and helium comparisons. The four-input common-Dirac semileptonic member proves positivity, an exact electron threshold and complete nucleon energy-window bounds. Native currents give three-dimensional beta parity, controlled massive leptonic decay and a tau leptonic-width ratio compatible with its unused comparison. A calibrated neutral tensor gives vacuum and matter records. Spin, magnetic response, Coulomb/Maxwell structure and atomic levels have explicit finite source domains. A full native kernel executes inside OpenWave, and two native clocks converge to a common weak-probe reading. A transverse Maxwell field shares finite bending, delay and redshift records with native clocks, while its tested physical-unit attachment fails. A full-Haar pure-gauge source admits a uniform-volume strong-coupling gap and local thermodynamic dynamics at fixed spacing. Its Casimir stability map extends to compact semisimple groups, without a nontrivial spatial-continuum identification. A resolved static charge probe joins the complete spatial core to its scale-dependent response; its vacuum-subtracted energies are reference-bound sector energies, without a physical-pole identification. These results use complete outputs and finite theorem reductions; their physical interpretations retain the fitted inputs, source distinctions and failed controls.

Controlled free-field limits, finite interacting responses and a conditional Einstein implication extend the same program toward physics in $3 + 1$ dimensions. A single particle comparison needs a common prepared source and clock, a persistent observable excitation, its physical symmetry and detector identification, and controlled volume and cutoff errors. Two neutral rest energies with a common relativistic dispersion determine a mass ratio without an absolute time calibration. Neither the complete species catalogue nor Einstein dynamics is a prerequisite for that comparison. Full physical composition additionally requires an interacting chiral continuum with the observed massive and massless sectors, and geometry responding to the stress and entropy of those same states. Finite symmetry and topology alone identify neither the measured pole masses nor that responsive geometry.

The scope of OPH is literal: a complete account would explain every public physical fact from one recoverable process, and would make its claims about existence and experience precise at the level they consume. The finite results give that ambition mathematical content; the realization maps and frozen tests state how the larger claim can succeed or fail. Physical adequacy depends on common-source constructions and independent measurements.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 3]. The paper release manifest records the release identifier (r2067) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [4]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [5, 6, 7, 10, 12, 72], a relationship disclosed here so that overlap

is visible.

Statements and Declarations

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Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the constructions, proofs, and counterexamples for the profile partition and structural characterizations, exact behavior-cut recovery, exact finite-state profile transport, adaptive repair stratification, the cumulative attempt-capacity classification, fixed computation-federation progress, and the fixed-federation execution classification, together with the mechanism-variant comparison kernel and conditional example and the source-derived phase-effect selection. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Daniel Ari Friedman contributed independent red-team audits of the papers, the book, the claim and premise ledgers, the Lean library, continuous integration and the custody records, which led to scoped headline claims, displayed values matched to their certified receipts and repaired falsification rules, and the independently reviewed programme-obstruction audit whose controls bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch.

All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

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Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 3]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 3]. The computational claims in this manuscript identify their principal files in Section 14.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [115, 117, 113, 114].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [109, 111, 118, 116, 112, 113, 114].
- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [110, 116, 113, 108].

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